

1 Instances (CPO)

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
1 20 2 10 rep 1.txt	20	2	10	Solution	30.08	721	639.00	11.37
2 20 2 10 rep 2.txt	20	2	10	Solution	30.01	610	580.00	4.92
3 20 2 10 rep 3.txt	20	2	10	Solution	30.01	553	513.00	7.23
4 20 2 10 rep 4.txt	20	2	10	Solution	30.01	794	706.00	11.08
5 20 2 10 rep 5.txt	20	2	10	Solution	30.01	561	541.00	3.57
6 20 2 10 rep 6.txt	20	2	10	Solution	30.01	648	600.00	7.41
7 20 2 10 rep 7.txt	20	2	10	Solution	30.01	620	584.00	5.81
8 20 2 10 rep 8.txt	20	2	10	Solution	30.01	946	782.00	17.34
9 20 2 10 rep 9.txt	20	2	10	Optimal	12.48	548	548.00	0.00
10 20 2 10 rep 10.txt	20	2	10	Solution	30.01	745	657.00	11.81
11 20 2 20 rep 1.txt	20	2	20	Optimal	11.54	323	323.00	0.00
12 20 2 20 rep 2.txt	20	2	20	Optimal	7.93	320	320.00	0.00
13 20 2 20 rep 3.txt	20	2	20	Optimal	10.20	289	289.00	0.00
14 20 2 20 rep 4.txt	20	2	20	Optimal	3.17	299	299.00	0.00
15 20 2 20 rep 5.txt	20	2	20	Optimal	1.13	320	320.00	0.00
16 20 2 20 rep 6.txt	20	2	20	Optimal	2.47	327	327.00	0.00
17 20 2 20 rep 7.txt	20	2	20	Optimal	5.86	352	352.00	0.00
18 20 2 20 rep 8.txt	20	2	20	Optimal	8.50	344	344.00	0.00
19 20 2 20 rep 9.txt	20	2	20	Optimal	0.93	290	289.00	0.34
20 20 2 20 rep 10.txt	20	2	20	Optimal	17.59	355	355.00	0.00
21 20 2 30 rep 1.txt	20	2	30	Optimal	0.05	185	185.00	0.00
22 20 2 30 rep 2.txt	20	2	30	Optimal	0.02	164	164.00	0.00
23 20 2 30 rep 3.txt	20	2	30	Optimal	0.20	231	231.00	0.00
24 20 2 30 rep 4.txt	20	2	30	Optimal	1.40	255	255.00	0.00
25 20 2 30 rep 5.txt	20	2	30	Optimal	0.70	235	235.00	0.00
26 20 2 30 rep 6.txt	20	2	30	Optimal	0.90	253	253.00	0.00
27 20 2 30 rep 7.txt	20	2	30	Optimal	2.75	209	209.00	0.00
28 20 2 30 rep 8.txt	20	2	30	Optimal	0.36	228	228.00	0.00
29 20 2 30 rep 9.txt	20	2	30	Optimal	3.38	234	234.00	0.00
30 20 2 30 rep 10.txt	20	2	30	Optimal	28.82	337	337.00	0.00
31 20 2 50 rep 1.txt	20	2	50	Optimal	0.02	191	191.00	0.00
32 20 2 50 rep 2.txt	20	2	50	Optimal	0.02	172	172.00	0.00
33 20 2 50 rep 3.txt	20	2	50	Optimal	0.06	173	173.00	0.00
34 20 2 50 rep 4.txt	20	2	50	Optimal	0.02	184	184.00	0.00
35 20 2 50 rep 5.txt	20	2	50	Optimal	0.02	179	179.00	0.00
36 20 2 50 rep 6.txt	20	2	50	Optimal	0.28	220	219.00	0.45
37 20 2 50 rep 7.txt	20	2	50	Optimal	0.02	170	170.00	0.00
38 20 2 50 rep 8.txt	20	2	50	Optimal	0.02	192	192.00	0.00
39 20 2 50 rep 9.txt	20	2	50	Optimal	0.11	204	204.00	0.00
40 20 2 50 rep 10.txt	20	2	50	Optimal	0.24	217	217.00	0.00
41 20 3 10 rep 1.txt	20	3	10	Solution	30.01	729	686.00	5.90
42 20 3 10 rep 2.txt	20	3	10	Solution	30.01	722	671.00	7.06

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Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
43 20 3 10 rep 3.txt	20	3	10	Solution	30.01	659	622.00	5.61
44 20 3 10 rep 4.txt	20	3	10	Solution	30.01	1104	847.00	23.28
45 20 3 10 rep 5.txt	20	3	10	Solution	30.01	726	697.00	3.99
46 20 3 10 rep 6.txt	20	3	10	Solution	30.01	888	756.00	14.86
47 20 3 10 rep 7.txt	20	3	10	Solution	30.01	832	778.00	6.49
48 20 3 10 rep 8.txt	20	3	10	Solution	30.01	739	683.00	7.58
49 20 3 10 rep 9.txt	20	3	10	Solution	30.02	876	752.00	14.16
50 20 3 10 rep 10.txt	20	3	10	Solution	30.01	925	811.00	12.32
51 20 3 20 rep 1.txt	20	3	20	Solution	30.01	449	439.00	2.23
52 20 3 20 rep 2.txt	20	3	20	Solution	30.01	382	367.00	3.93
53 20 3 20 rep 3.txt	20	3	20	Solution	30.01	385	374.00	2.86
54 20 3 20 rep 4.txt	20	3	20	Optimal	2.75	380	380.00	0.00
55 20 3 20 rep 5.txt	20	3	20	Solution	30.01	402	382.00	4.98
56 20 3 20 rep 6.txt	20	3	20	Optimal	15.66	350	350.00	0.00
57 20 3 20 rep 7.txt	20	3	20	Optimal	5.80	431	431.00	0.00
58 20 3 20 rep 8.txt	20	3	20	Optimal	10.77	331	331.00	0.00
59 20 3 20 rep 9.txt	20	3	20	Solution	30.01	358	353.00	1.40
60 20 3 20 rep 10.txt	20	3	20	Optimal	7.00	365	365.00	0.00
61 20 3 30 rep 1.txt	20	3	30	Optimal	9.29	338	338.00	0.00
62 20 3 30 rep 2.txt	20	3	30	Optimal	0.73	241	241.00	0.00
63 20 3 30 rep 3.txt	20	3	30	Optimal	0.16	269	269.00	0.00
64 20 3 30 rep 4.txt	20	3	30	Optimal	0.97	304	304.00	0.00
65 20 3 30 rep 5.txt	20	3	30	Optimal	1.79	263	263.00	0.00
66 20 3 30 rep 6.txt	20	3	30	Optimal	0.25	259	259.00	0.00
67 20 3 30 rep 7.txt	20	3	30	Optimal	8.09	377	377.00	0.00
68 20 3 30 rep 8.txt	20	3	30	Optimal	1.02	287	287.00	0.00
69 20 3 30 rep 9.txt	20	3	30	Optimal	0.40	241	241.00	0.00
70 20 3 30 rep 10.txt	20	3	30	Optimal	0.18	228	228.00	0.00
71 20 3 50 rep 1.txt	20	3	50	Optimal	0.02	259	259.00	0.00
72 20 3 50 rep 2.txt	20	3	50	Optimal	0.02	233	233.00	0.00
73 20 3 50 rep 3.txt	20	3	50	Optimal	0.02	234	234.00	0.00
74 20 3 50 rep 4.txt	20	3	50	Optimal	0.02	270	270.00	0.00
75 20 3 50 rep 5.txt	20	3	50	Optimal	0.02	247	247.00	0.00
76 20 3 50 rep 6.txt	20	3	50	Optimal	0.11	245	245.00	0.00
77 20 3 50 rep 7.txt	20	3	50	Optimal	0.11	217	217.00	0.00
78 20 3 50 rep 8.txt	20	3	50	Optimal	0.02	232	232.00	0.00
79 20 3 50 rep 9.txt	20	3	50	Optimal	0.26	241	241.00	0.00
80 20 3 50 rep 10.txt	20	3	50	Optimal	0.02	223	223.00	0.00
81 20 5 10 rep 1.txt	20	5	10	Solution	30.01	727	594.00	18.29
82 20 5 10 rep 2.txt	20	5	10	Solution	30.01	926	802.00	13.39
83 20 5 10 rep 3.txt	20	5	10	Solution	30.01	829	743.00	10.37
84 20 5 10 rep 4.txt	20	5	10	Solution	30.01	783	687.00	12.26
85 20 5 10 rep 5.txt	20	5	10	Solution	30.01	774	703.00	9.17
86 20 5 10 rep 6.txt	20	5	10	Solution	30.01	995	883.00	11.26
87 20 5 10 rep 7.txt	20	5	10	Solution	30.01	758	696.00	8.18

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Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
88 20 5 10 rep 8.txt	20	5	10	Solution	30.01	925	790.00	14.59
89 20 5 10 rep 9.txt	20	5	10	Solution	30.01	881	803.00	8.85
90 20 5 10 rep 10.txt	20	5	10	Solution	30.01	931	901.00	3.22
91 20 5 20 rep 1.txt	20	5	20	Solution	30.01	486	459.00	5.56
92 20 5 20 rep 2.txt	20	5	20	Solution	30.01	501	482.00	3.79
93 20 5 20 rep 3.txt	20	5	20	Solution	30.01	517	484.00	6.38
94 20 5 20 rep 4.txt	20	5	20	Optimal	5.09	447	447.00	0.00
95 20 5 20 rep 5.txt	20	5	20	Solution	30.01	548	502.00	8.39
96 20 5 20 rep 6.txt	20	5	20	Solution	30.01	495	461.00	6.87
97 20 5 20 rep 7.txt	20	5	20	Solution	30.01	548	530.00	3.28
98 20 5 20 rep 8.txt	20	5	20	Solution	30.00	483	447.00	7.45
99 20 5 20 rep 9.txt	20	5	20	Solution	30.01	463	418.00	9.72
100 20 5 20 rep 10.txt	20	5	20	Solution	30.01	495	475.00	4.04
101 20 5 30 rep 1.txt	20	5	30	Optimal	0.36	370	370.00	0.00
102 20 5 30 rep 2.txt	20	5	30	Solution	30.01	389	371.00	4.63
103 20 5 30 rep 3.txt	20	5	30	Optimal	0.19	371	371.00	0.00
104 20 5 30 rep 4.txt	20	5	30	Optimal	1.22	400	400.00	0.00
105 20 5 30 rep 5.txt	20	5	30	Optimal	10.66	405	405.00	0.00
106 20 5 30 rep 6.txt	20	5	30	Optimal	0.11	388	388.00	0.00
107 20 5 30 rep 7.txt	20	5	30	Optimal	0.11	424	424.00	0.00
108 20 5 30 rep 8.txt	20	5	30	Optimal	0.02	407	407.00	0.00
109 20 5 30 rep 9.txt	20	5	30	Optimal	0.35	358	358.00	0.00
110 20 5 30 rep 10.txt	20	5	30	Optimal	0.05	427	427.00	0.00
111 20 5 50 rep 1.txt	20	5	50	Optimal	0.02	408	408.00	0.00
112 20 5 50 rep 2.txt	20	5	50	Optimal	0.02	418	418.00	0.00
113 20 5 50 rep 3.txt	20	5	50	Optimal	0.02	357	357.00	0.00
114 20 5 50 rep 4.txt	20	5	50	Optimal	0.02	428	428.00	0.00
115 20 5 50 rep 5.txt	20	5	50	Optimal	0.02	373	373.00	0.00
116 20 5 50 rep 6.txt	20	5	50	Optimal	0.02	365	365.00	0.00
117 20 5 50 rep 7.txt	20	5	50	Optimal	0.02	387	387.00	0.00
118 20 5 50 rep 8.txt	20	5	50	Optimal	0.02	348	348.00	0.00
119 20 5 50 rep 9.txt	20	5	50	Optimal	0.02	405	405.00	0.00
120 20 5 50 rep 10.txt	20	5	50	Optimal	0.02	417	417.00	0.00
121 20 10 10 rep 1.txt	20	10	10	Solution	30.01	1148	1003.00	12.63
122 20 10 10 rep 2.txt	20	10	10	Solution	30.01	1276	1013.00	20.61
123 20 10 10 rep 3.txt	20	10	10	Solution	30.01	1229	1063.00	13.51
124 20 10 10 rep 4.txt	20	10	10	Solution	30.01	1268	992.00	21.77
125 20 10 10 rep 5.txt	20	10	10	Solution	30.01	1279	1066.00	16.65
126 20 10 10 rep 6.txt	20	10	10	Solution	30.01	1187	1035.00	12.81
127 20 10 10 rep 7.txt	20	10	10	Solution	30.02	1161	1004.00	13.52
128 20 10 10 rep 8.txt	20	10	10	Solution	30.01	1426	1125.00	21.11
129 20 10 10 rep 9.txt	20	10	10	Solution	30.01	1191	1030.00	13.52
130 20 10 10 rep 10.txt	20	10	10	Solution	30.01	1291	1060.00	17.89
131 20 10 20 rep 1.txt	20	10	20	Solution	30.01	745	674.00	9.53
132 20 10 20 rep 2.txt	20	10	20	Solution	30.01	797	746.00	6.40

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Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
133 20 10 20 rep 3.txt	20	10	20	Solution	30.01	756	705.00	6.75
134 20 10 20 rep 4.txt	20	10	20	Solution	30.01	773	676.00	12.55
135 20 10 20 rep 5.txt	20	10	20	Solution	30.01	847	771.00	8.97
136 20 10 20 rep 6.txt	20	10	20	Solution	30.01	826	739.00	10.53
137 20 10 20 rep 7.txt	20	10	20	Solution	30.01	810	742.00	8.40
138 20 10 20 rep 8.txt	20	10	20	Solution	30.02	752	669.00	11.04
139 20 10 20 rep 9.txt	20	10	20	Solution	30.02	838	760.00	9.31
140 20 10 20 rep 10.txt	20	10	20	Solution	30.01	773	733.00	5.17
141 20 10 30 rep 1.txt	20	10	30	Optimal	11.82	713	713.00	0.00
142 20 10 30 rep 2.txt	20	10	30	Optimal	0.02	720	720.00	0.00
143 20 10 30 rep 3.txt	20	10	30	Optimal	0.27	682	682.00	0.00
144 20 10 30 rep 4.txt	20	10	30	Optimal	2.87	642	642.00	0.00
145 20 10 30 rep 5.txt	20	10	30	Optimal	3.25	697	697.00	0.00
146 20 10 30 rep 6.txt	20	10	30	Optimal	18.17	616	616.00	0.00
147 20 10 30 rep 7.txt	20	10	30	Solution	30.01	650	641.00	1.38
148 20 10 30 rep 8.txt	20	10	30	Optimal	0.11	631	631.00	0.00
149 20 10 30 rep 9.txt	20	10	30	Optimal	12.89	680	680.00	0.00
150 20 10 30 rep 10.txt	20	10	30	Optimal	20.61	692	692.00	0.00
151 20 10 50 rep 1.txt	20	10	50	Optimal	0.03	711	711.00	0.00
152 20 10 50 rep 2.txt	20	10	50	Optimal	0.02	621	621.00	0.00
153 20 10 50 rep 3.txt	20	10	50	Optimal	0.02	669	669.00	0.00
154 20 10 50 rep 4.txt	20	10	50	Optimal	0.02	740	740.00	0.00
155 20 10 50 rep 5.txt	20	10	50	Optimal	0.02	686	686.00	0.00
156 20 10 50 rep 6.txt	20	10	50	Optimal	0.02	715	715.00	0.00
157 20 10 50 rep 7.txt	20	10	50	Optimal	0.02	612	612.00	0.00
158 20 10 50 rep 8.txt	20	10	50	Optimal	0.02	628	628.00	0.00
159 20 10 50 rep 9.txt	20	10	50	Optimal	0.02	744	744.00	0.00
160 20 10 50 rep 10.txt	20	10	50	Optimal	0.02	707	707.00	0.00
161 30 2 10 rep 1.txt	30	2	10	Solution	30.01	1255	1072.00	14.58
162 30 2 10 rep 2.txt	30	2	10	Solution	30.01	826	804.00	2.66
163 30 2 10 rep 3.txt	30	2	10	Solution	30.01	863	837.00	3.01
164 30 2 10 rep 4.txt	30	2	10	Solution	30.01	1078	1006.00	6.68
165 30 2 10 rep 5.txt	30	2	10	Solution	30.01	1177	1027.00	12.74
166 30 2 10 rep 6.txt	30	2	10	Solution	30.01	846	832.00	1.65
167 30 2 10 rep 7.txt	30	2	10	Solution	30.01	719	704.00	2.09
168 30 2 10 rep 8.txt	30	2	10	Solution	30.01	890	852.00	4.27
169 30 2 10 rep 9.txt	30	2	10	Solution	30.01	983	895.00	8.95
170 30 2 10 rep 10.txt	30	2	10	Solution	30.01	1140	1093.00	4.12
171 30 2 20 rep 1.txt	30	2	20	Solution	30.01	427	416.00	2.58
172 30 2 20 rep 2.txt	30	2	20	Solution	30.01	515	506.00	1.75
173 30 2 20 rep 3.txt	30	2	20	Solution	30.01	405	397.00	1.98
174 30 2 20 rep 4.txt	30	2	20	Solution	30.01	420	418.00	0.48
175 30 2 20 rep 5.txt	30	2	20	Solution	30.01	446	441.00	1.12
176 30 2 20 rep 6.txt	30	2	20	Solution	30.01	436	427.00	2.06
177 30 2 20 rep 7.txt	30	2	20	Solution	30.01	492	483.00	1.83

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
178 30 2 20 rep 8.txt	30	2	20	Solution	30.01	387	378.00	2.33
179 30 2 20 rep 9.txt	30	2	20	Solution	30.01	437	427.00	2.29
180 30 2 20 rep 10.txt	30	2	20	Solution	30.01	545	516.00	5.32
181 30 2 30 rep 1.txt	30	2	30	Solution	30.01	311	309.00	0.64
182 30 2 30 rep 2.txt	30	2	30	Optimal	9.58	229	229.00	0.00
183 30 2 30 rep 3.txt	30	2	30	Solution	30.01	332	329.00	0.90
184 30 2 30 rep 4.txt	30	2	30	Solution	30.01	327	324.00	0.92
185 30 2 30 rep 5.txt	30	2	30	Solution	30.01	290	286.00	1.38
186 30 2 30 rep 6.txt	30	2	30	Solution	30.01	338	332.00	1.78
187 30 2 30 rep 7.txt	30	2	30	Solution	30.01	373	367.00	1.61
188 30 2 30 rep 8.txt	30	2	30	Solution	30.01	348	345.00	0.86
189 30 2 30 rep 9.txt	30	2	30	Solution	30.01	400	392.00	2.00
190 30 2 30 rep 10.txt	30	2	30	Solution	30.01	312	307.00	1.60
191 30 2 50 rep 1.txt	30	2	50	Solution	30.01	204	201.00	1.47
192 30 2 50 rep 2.txt	30	2	50	Optimal	0.93	195	195.00	0.00
193 30 2 50 rep 3.txt	30	2	50	Solution	30.01	205	201.00	1.95
194 30 2 50 rep 4.txt	30	2	50	Optimal	5.33	201	201.00	0.00
195 30 2 50 rep 5.txt	30	2	50	Optimal	0.14	195	195.00	0.00
196 30 2 50 rep 6.txt	30	2	50	Solution	30.01	220	216.00	1.82
197 30 2 50 rep 7.txt	30	2	50	Solution	30.01	243	241.00	0.82
198 30 2 50 rep 8.txt	30	2	50	Optimal	0.02	176	176.00	0.00
199 30 2 50 rep 9.txt	30	2	50	Optimal	0.85	174	174.00	0.00
200 30 2 50 rep 10.txt	30	2	50	Optimal	2.34	229	229.00	0.00
201 30 3 10 rep 1.txt	30	3	10	Solution	30.01	1069	909.00	14.97
202 30 3 10 rep 2.txt	30	3	10	Solution	30.01	1062	966.00	9.04
203 30 3 10 rep 3.txt	30	3	10	Solution	30.01	1165	1080.00	7.30
204 30 3 10 rep 4.txt	30	3	10	Solution	30.01	870	847.00	2.64
205 30 3 10 rep 5.txt	30	3	10	Solution	30.01	1059	1015.00	4.15
206 30 3 10 rep 6.txt	30	3	10	Solution	30.01	1025	867.00	15.41
207 30 3 10 rep 7.txt	30	3	10	Solution	30.01	923	854.00	7.48
208 30 3 10 rep 8.txt	30	3	10	Solution	30.00	818	789.00	3.55
209 30 3 10 rep 9.txt	30	3	10	Solution	30.01	1107	990.00	10.57
210 30 3 10 rep 10.txt	30	3	10	Solution	30.01	975	938.00	3.79
211 30 3 20 rep 1.txt	30	3	20	Solution	30.01	524	516.00	1.53
212 30 3 20 rep 2.txt	30	3	20	Solution	30.01	599	586.00	2.17
213 30 3 20 rep 3.txt	30	3	20	Solution	30.01	470	447.00	4.89
214 30 3 20 rep 4.txt	30	3	20	Solution	30.01	597	576.00	3.52
215 30 3 20 rep 5.txt	30	3	20	Solution	30.01	546	518.00	5.13
216 30 3 20 rep 6.txt	30	3	20	Solution	30.01	588	561.00	4.59
217 30 3 20 rep 7.txt	30	3	20	Solution	30.01	633	595.00	6.00
218 30 3 20 rep 8.txt	30	3	20	Solution	30.01	555	540.00	2.70
219 30 3 20 rep 9.txt	30	3	20	Solution	30.01	528	512.00	3.03
220 30 3 20 rep 10.txt	30	3	20	Solution	30.01	518	495.00	4.44
221 30 3 30 rep 1.txt	30	3	30	Solution	30.01	372	367.00	1.34
222 30 3 30 rep 2.txt	30	3	30	Solution	30.01	340	325.00	4.41

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223 30 3 30 rep 3.txt	30	3	30	Solution	30.01	428	420.00	1.87
224 30 3 30 rep 4.txt	30	3	30	Solution	30.01	377	366.00	2.92
225 30 3 30 rep 5.txt	30	3	30	Solution	30.01	309	301.00	2.59
226 30 3 30 rep 6.txt	30	3	30	Solution	30.01	365	361.00	1.10
227 30 3 30 rep 7.txt	30	3	30	Solution	30.01	393	387.00	1.53
228 30 3 30 rep 8.txt	30	3	30	Solution	30.01	365	349.00	4.38
229 30 3 30 rep 9.txt	30	3	30	Solution	30.01	350	346.00	1.14
230 30 3 30 rep 10.txt	30	3	30	Solution	30.01	338	317.00	6.21
231 30 3 50 rep 1.txt	30	3	50	Optimal	1.27	259	259.00	0.00
232 30 3 50 rep 2.txt	30	3	50	Optimal	0.85	238	238.00	0.00
233 30 3 50 rep 3.txt	30	3	50	Solution	30.01	260	257.00	1.15
234 30 3 50 rep 4.txt	30	3	50	Optimal	10.95	267	267.00	0.00
235 30 3 50 rep 5.txt	30	3	50	Optimal	0.03	256	256.00	0.00
236 30 3 50 rep 6.txt	30	3	50	Optimal	0.02	260	260.00	0.00
237 30 3 50 rep 7.txt	30	3	50	Optimal	4.52	266	266.00	0.00
238 30 3 50 rep 8.txt	30	3	50	Optimal	1.05	261	261.00	0.00
239 30 3 50 rep 9.txt	30	3	50	Optimal	8.38	236	236.00	0.00
240 30 3 50 rep 10.txt	30	3	50	Optimal	0.05	263	263.00	0.00
241 30 5 10 rep 1.txt	30	5	10	Solution	30.01	1147	1080.00	5.84
242 30 5 10 rep 2.txt	30	5	10	Solution	30.01	1097	925.00	15.68
243 30 5 10 rep 3.txt	30	5	10	Solution	30.01	1177	1047.00	11.05
244 30 5 10 rep 4.txt	30	5	10	Solution	30.01	1052	1004.00	4.56
245 30 5 10 rep 5.txt	30	5	10	Solution	30.01	1111	997.00	10.26
246 30 5 10 rep 6.txt	30	5	10	Solution	30.01	1295	1165.00	10.04
247 30 5 10 rep 7.txt	30	5	10	Solution	30.01	1130	1020.00	9.73
248 30 5 10 rep 8.txt	30	5	10	Solution	30.01	995	908.00	8.74
249 30 5 10 rep 9.txt	30	5	10	Solution	30.01	1072	965.00	9.98
250 30 5 10 rep 10.txt	30	5	10	Solution	30.01	1176	1136.00	3.40
251 30 5 20 rep 1.txt	30	5	20	Solution	30.02	658	625.00	5.02
252 30 5 20 rep 2.txt	30	5	20	Solution	30.01	688	659.00	4.22
253 30 5 20 rep 3.txt	30	5	20	Solution	30.01	616	562.00	8.77
254 30 5 20 rep 4.txt	30	5	20	Solution	30.01	637	610.00	4.24
255 30 5 20 rep 5.txt	30	5	20	Solution	30.01	600	558.00	7.00
256 30 5 20 rep 6.txt	30	5	20	Solution	30.01	606	566.00	6.60
257 30 5 20 rep 7.txt	30	5	20	Solution	30.01	679	596.00	12.22
258 30 5 20 rep 8.txt	30	5	20	Solution	30.01	604	560.00	7.28
259 30 5 20 rep 9.txt	30	5	20	Solution	30.01	683	621.00	9.08
260 30 5 20 rep 10.txt	30	5	20	Solution	30.01	679	650.00	4.27
261 30 5 30 rep 1.txt	30	5	30	Solution	30.01	429	415.00	3.26
262 30 5 30 rep 2.txt	30	5	30	Solution	30.01	462	419.00	9.31
263 30 5 30 rep 3.txt	30	5	30	Solution	30.01	466	442.00	5.15
264 30 5 30 rep 4.txt	30	5	30	Solution	30.01	442	441.00	0.23
265 30 5 30 rep 5.txt	30	5	30	Solution	30.01	453	426.00	5.96
266 30 5 30 rep 6.txt	30	5	30	Solution	30.01	487	446.00	8.42
267 30 5 30 rep 7.txt	30	5	30	Solution	30.01	469	443.00	5.54

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
268 30 5 30 rep 8.txt	30	5	30	Solution	30.01	519	491.00	5.39
269 30 5 30 rep 9.txt	30	5	30	Solution	30.01	474	458.00	3.38
270 30 5 30 rep 10.txt	30	5	30	Solution	30.01	501	489.00	2.40
271 30 5 50 rep 1.txt	30	5	50	Solution	30.01	401	393.00	2.00
272 30 5 50 rep 2.txt	30	5	50	Optimal	0.09	367	367.00	0.00
273 30 5 50 rep 3.txt	30	5	50	Optimal	0.04	397	397.00	0.00
274 30 5 50 rep 4.txt	30	5	50	Optimal	0.19	411	411.00	0.00
275 30 5 50 rep 5.txt	30	5	50	Optimal	0.05	399	399.00	0.00
276 30 5 50 rep 6.txt	30	5	50	Optimal	0.03	392	392.00	0.00
277 30 5 50 rep 7.txt	30	5	50	Optimal	0.03	406	406.00	0.00
278 30 5 50 rep 8.txt	30	5	50	Solution	30.01	401	396.00	1.25
279 30 5 50 rep 9.txt	30	5	50	Optimal	0.02	352	352.00	0.00
280 30 5 50 rep 10.txt	30	5	50	Optimal	0.14	399	399.00	0.00
281 30 10 10 rep 1.txt	30	10	10	Solution	30.02	1529	1303.00	14.78
282 30 10 10 rep 2.txt	30	10	10	Solution	30.02	1637	1273.00	22.24
283 30 10 10 rep 3.txt	30	10	10	Solution	30.01	1398	1212.00	13.30
284 30 10 10 rep 4.txt	30	10	10	Solution	30.01	1454	1201.00	17.40
285 30 10 10 rep 5.txt	30	10	10	Solution	30.01	1550	1269.00	18.13
286 30 10 10 rep 6.txt	30	10	10	Solution	30.02	1563	1302.00	16.70
287 30 10 10 rep 7.txt	30	10	10	Solution	30.01	1503	1214.00	19.23
288 30 10 10 rep 8.txt	30	10	10	Solution	30.02	1450	1242.00	14.34
289 30 10 10 rep 9.txt	30	10	10	Solution	30.01	1612	1353.00	16.07
290 30 10 10 rep 10.txt	30	10	10	Solution	30.01	1589	1259.00	20.77
291 30 10 20 rep 1.txt	30	10	20	Solution	30.02	893	770.00	13.77
292 30 10 20 rep 2.txt	30	10	20	Solution	30.01	957	845.00	11.70
293 30 10 20 rep 3.txt	30	10	20	Solution	30.01	909	767.00	15.62
294 30 10 20 rep 4.txt	30	10	20	Solution	30.01	857	741.00	13.54
295 30 10 20 rep 5.txt	30	10	20	Solution	30.01	962	814.00	15.38
296 30 10 20 rep 6.txt	30	10	20	Solution	30.01	936	859.00	8.23
297 30 10 20 rep 7.txt	30	10	20	Solution	30.01	877	794.00	9.46
298 30 10 20 rep 8.txt	30	10	20	Solution	30.02	1010	847.00	16.14
299 30 10 20 rep 9.txt	30	10	20	Solution	30.03	888	750.00	15.54
300 30 10 20 rep 10.txt	30	10	20	Solution	30.01	909	792.00	12.87
301 30 10 30 rep 1.txt	30	10	30	Solution	30.03	739	712.00	3.65
302 30 10 30 rep 2.txt	30	10	30	Solution	30.03	764	704.00	7.85
303 30 10 30 rep 3.txt	30	10	30	Solution	30.03	733	679.00	7.37
304 30 10 30 rep 4.txt	30	10	30	Solution	30.03	707	674.00	4.67
305 30 10 30 rep 5.txt	30	10	30	Optimal	0.62	777	777.00	0.00
306 30 10 30 rep 6.txt	30	10	30	Solution	30.03	738	717.00	2.85
307 30 10 30 rep 7.txt	30	10	30	Solution	30.03	749	676.00	9.75
308 30 10 30 rep 8.txt	30	10	30	Solution	30.03	734	670.00	8.72
309 30 10 30 rep 9.txt	30	10	30	Solution	30.03	749	702.00	6.28
310 30 10 30 rep 10.txt	30	10	30	Solution	30.02	710	695.00	2.11
311 30 10 50 rep 1.txt	30	10	50	Optimal	0.05	760	760.00	0.00
312 30 10 50 rep 2.txt	30	10	50	Optimal	0.05	694	694.00	0.00

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
313 30 10 50 rep 3.txt	30	10	50	Optimal	0.06	762	762.00	0.00
314 30 10 50 rep 4.txt	30	10	50	Optimal	0.03	671	671.00	0.00
315 30 10 50 rep 5.txt	30	10	50	Optimal	0.03	747	747.00	0.00
316 30 10 50 rep 6.txt	30	10	50	Solution	30.01	667	653.00	2.10
317 30 10 50 rep 7.txt	30	10	50	Optimal	0.26	659	659.00	0.00
318 30 10 50 rep 8.txt	30	10	50	Optimal	0.30	668	668.00	0.00
319 30 10 50 rep 9.txt	30	10	50	Optimal	0.28	658	658.00	0.00
320 30 10 50 rep 10.txt	30	10	50	Optimal	0.02	695	695.00	0.00
321 50 2 10 rep 1.txt	50	2	10	Solution	30.02	1594	1505.00	5.58
322 50 2 10 rep 2.txt	50	2	10	Solution	30.03	1596	1450.00	9.15
323 50 2 10 rep 3.txt	50	2	10	Solution	30.01	1622	1505.00	7.21
324 50 2 10 rep 4.txt	50	2	10	Solution	30.03	1819	1689.00	7.15
325 50 2 10 rep 5.txt	50	2	10	Solution	30.02	1676	1648.00	1.67
326 50 2 10 rep 6.txt	50	2	10	Solution	30.01	1290	1277.00	1.01
327 50 2 10 rep 7.txt	50	2	10	Solution	30.01	1747	1560.00	10.70
328 50 2 10 rep 8.txt	50	2	10	Solution	30.01	1611	1517.00	5.83
329 50 2 10 rep 9.txt	50	2	10	Solution	30.01	1373	1310.00	4.59
330 50 2 10 rep 10.txt	50	2	10	Solution	30.02	1499	1401.00	6.54
331 50 2 20 rep 1.txt	50	2	20	Solution	30.02	849	835.00	1.65
332 50 2 20 rep 2.txt	50	2	20	Solution	30.03	654	644.00	1.53
333 50 2 20 rep 3.txt	50	2	20	Solution	30.01	773	765.00	1.03
334 50 2 20 rep 4.txt	50	2	20	Solution	30.03	672	663.00	1.34
335 50 2 20 rep 5.txt	50	2	20	Solution	30.01	816	804.00	1.47
336 50 2 20 rep 6.txt	50	2	20	Solution	30.02	811	803.00	0.99
337 50 2 20 rep 7.txt	50	2	20	Solution	30.01	678	661.00	2.51
338 50 2 20 rep 8.txt	50	2	20	Solution	30.01	792	765.00	3.41
339 50 2 20 rep 9.txt	50	2	20	Solution	30.03	741	732.00	1.21
340 50 2 20 rep 10.txt	50	2	20	Solution	30.03	678	671.00	1.03
341 50 2 30 rep 1.txt	50	2	30	Solution	30.02	462	450.00	2.60
342 50 2 30 rep 2.txt	50	2	30	Solution	30.00	476	468.00	1.68
343 50 2 30 rep 3.txt	50	2	30	Solution	30.01	473	461.00	2.54
344 50 2 30 rep 4.txt	50	2	30	Solution	30.01	506	502.00	0.79
345 50 2 30 rep 5.txt	50	2	30	Solution	30.02	600	586.00	2.33
346 50 2 30 rep 6.txt	50	2	30	Solution	30.01	538	528.00	1.86
347 50 2 30 rep 7.txt	50	2	30	Solution	30.02	555	543.00	2.16
348 50 2 30 rep 8.txt	50	2	30	Solution	30.01	525	512.00	2.48
349 50 2 30 rep 9.txt	50	2	30	Solution	30.02	478	464.00	2.93
350 50 2 30 rep 10.txt	50	2	30	Solution	30.00	435	423.00	2.76
351 50 2 50 rep 1.txt	50	2	50	Solution	30.03	333	330.00	0.90
352 50 2 50 rep 2.txt	50	2	50	Solution	30.05	287	281.00	2.09
353 50 2 50 rep 3.txt	50	2	50	Solution	30.01	399	393.00	1.50
354 50 2 50 rep 4.txt	50	2	50	Solution	30.01	339	334.00	1.47
355 50 2 50 rep 5.txt	50	2	50	Solution	30.00	298	295.00	1.01
356 50 2 50 rep 6.txt	50	2	50	Solution	30.01	335	331.00	1.19
357 50 2 50 rep 7.txt	50	2	50	Solution	30.03	319	315.00	1.25

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
358 50 2 50 rep 8.txt	50	2	50	Solution	30.02	324	322.00	0.62
359 50 2 50 rep 9.txt	50	2	50	Solution	30.02	331	323.00	2.42
360 50 2 50 rep 10.txt	50	2	50	Solution	30.01	281	279.00	0.71
361 50 3 10 rep 1.txt	50	3	10	Solution	30.03	1831	1710.00	6.61
362 50 3 10 rep 2.txt	50	3	10	Solution	30.03	1578	1490.00	5.58
363 50 3 10 rep 3.txt	50	3	10	Solution	30.02	1546	1428.00	7.63
364 50 3 10 rep 4.txt	50	3	10	Solution	30.02	1773	1655.00	6.66
365 50 3 10 rep 5.txt	50	3	10	Solution	30.02	1889	1747.00	7.52
366 50 3 10 rep 6.txt	50	3	10	Solution	30.01	1683	1551.00	7.84
367 50 3 10 rep 7.txt	50	3	10	Solution	30.01	1528	1438.00	5.89
368 50 3 10 rep 8.txt	50	3	10	Solution	30.02	1547	1476.00	4.59
369 50 3 10 rep 9.txt	50	3	10	Solution	30.03	1862	1714.00	7.95
370 50 3 10 rep 10.txt	50	3	10	Solution	30.02	1420	1321.00	6.97
371 50 3 20 rep 1.txt	50	3	20	Solution	30.02	749	722.00	3.60
372 50 3 20 rep 2.txt	50	3	20	Solution	30.01	815	757.00	7.12
373 50 3 20 rep 3.txt	50	3	20	Solution	30.02	786	760.00	3.31
374 50 3 20 rep 4.txt	50	3	20	Solution	30.02	796	775.00	2.64
375 50 3 20 rep 5.txt	50	3	20	Solution	30.02	760	736.00	3.16
376 50 3 20 rep 6.txt	50	3	20	Solution	30.02	874	860.00	1.60
377 50 3 20 rep 7.txt	50	3	20	Solution	30.02	810	799.00	1.36
378 50 3 20 rep 8.txt	50	3	20	Solution	30.01	867	841.00	3.00
379 50 3 20 rep 9.txt	50	3	20	Solution	30.01	948	925.00	2.43
380 50 3 20 rep 10.txt	50	3	20	Solution	30.02	800	753.00	5.88
381 50 3 30 rep 1.txt	50	3	30	Solution	30.02	522	516.00	1.15
382 50 3 30 rep 2.txt	50	3	30	Solution	30.02	503	477.00	5.17
383 50 3 30 rep 3.txt	50	3	30	Solution	30.01	503	488.00	2.98
384 50 3 30 rep 4.txt	50	3	30	Solution	30.02	559	539.00	3.58
385 50 3 30 rep 5.txt	50	3	30	Solution	30.01	538	512.00	4.83
386 50 3 30 rep 6.txt	50	3	30	Solution	30.02	589	559.00	5.09
387 50 3 30 rep 7.txt	50	3	30	Solution	30.03	507	488.00	3.75
388 50 3 30 rep 8.txt	50	3	30	Solution	30.02	598	586.00	2.01
389 50 3 30 rep 9.txt	50	3	30	Solution	30.02	516	492.00	4.65
390 50 3 30 rep 10.txt	50	3	30	Solution	30.02	526	515.00	2.09
391 50 3 50 rep 1.txt	50	3	50	Solution	30.02	336	316.00	5.95
392 50 3 50 rep 2.txt	50	3	50	Solution	30.01	294	286.00	2.72
393 50 3 50 rep 3.txt	50	3	50	Solution	30.02	368	362.00	1.63
394 50 3 50 rep 4.txt	50	3	50	Solution	30.02	364	352.00	3.30
395 50 3 50 rep 5.txt	50	3	50	Solution	30.01	377	361.00	4.24
396 50 3 50 rep 6.txt	50	3	50	Solution	30.01	353	348.00	1.42
397 50 3 50 rep 7.txt	50	3	50	Solution	30.02	366	345.00	5.74
398 50 3 50 rep 8.txt	50	3	50	Solution	30.02	356	346.00	2.81
399 50 3 50 rep 9.txt	50	3	50	Solution	30.01	325	322.00	0.92
400 50 3 50 rep 10.txt	50	3	50	Solution	30.01	345	325.00	5.80
401 50 5 10 rep 1.txt	50	5	10	Solution	30.02	1954	1841.00	5.78
402 50 5 10 rep 2.txt	50	5	10	Solution	30.03	1656	1425.00	13.95

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
403 50 5 10 rep 3.txt	50	5	10	Solution	30.03	1705	1501.00	11.96
404 50 5 10 rep 4.txt	50	5	10	Solution	30.03	1992	1837.00	7.78
405 50 5 10 rep 5.txt	50	5	10	Solution	30.04	1704	1498.00	12.09
406 50 5 10 rep 6.txt	50	5	10	Solution	30.02	1885	1729.00	8.28
407 50 5 10 rep 7.txt	50	5	10	Solution	30.03	1920	1783.00	7.14
408 50 5 10 rep 8.txt	50	5	10	Solution	30.03	1897	1764.00	7.01
409 50 5 10 rep 9.txt	50	5	10	Solution	30.03	1907	1767.00	7.34
410 50 5 10 rep 10.txt	50	5	10	Solution	30.02	1970	1767.00	10.30
411 50 5 20 rep 1.txt	50	5	20	Solution	30.02	899	820.00	8.79
412 50 5 20 rep 2.txt	50	5	20	Solution	30.02	901	869.00	3.55
413 50 5 20 rep 3.txt	50	5	20	Solution	30.02	872	795.00	8.83
414 50 5 20 rep 4.txt	50	5	20	Solution	30.03	914	880.00	3.72
415 50 5 20 rep 5.txt	50	5	20	Solution	30.03	875	798.00	8.80
416 50 5 20 rep 6.txt	50	5	20	Solution	30.02	782	710.00	9.21
417 50 5 20 rep 7.txt	50	5	20	Solution	30.02	921	849.00	7.82
418 50 5 20 rep 8.txt	50	5	20	Solution	30.03	849	806.00	5.06
419 50 5 20 rep 9.txt	50	5	20	Solution	30.02	868	823.00	5.18
420 50 5 20 rep 10.txt	50	5	20	Solution	30.04	857	779.00	9.10
421 50 5 30 rep 1.txt	50	5	30	Solution	30.02	670	605.00	9.70
422 50 5 30 rep 2.txt	50	5	30	Solution	30.02	706	659.00	6.66
423 50 5 30 rep 3.txt	50	5	30	Solution	30.02	680	660.00	2.94
424 50 5 30 rep 4.txt	50	5	30	Solution	30.02	593	576.00	2.87
425 50 5 30 rep 5.txt	50	5	30	Solution	30.02	646	628.00	2.79
426 50 5 30 rep 6.txt	50	5	30	Solution	30.02	650	613.00	5.69
427 50 5 30 rep 7.txt	50	5	30	Solution	30.03	657	624.00	5.02
428 50 5 30 rep 8.txt	50	5	30	Solution	30.02	725	699.00	3.59
429 50 5 30 rep 9.txt	50	5	30	Solution	30.02	645	612.00	5.12
430 50 5 30 rep 10.txt	50	5	30	Solution	30.02	677	660.00	2.51
431 50 5 50 rep 1.txt	50	5	50	Solution	30.02	440	416.00	5.45
432 50 5 50 rep 2.txt	50	5	50	Solution	30.02	466	449.00	3.65
433 50 5 50 rep 3.txt	50	5	50	Solution	30.01	461	445.00	3.47
434 50 5 50 rep 4.txt	50	5	50	Solution	30.02	480	449.00	6.46
435 50 5 50 rep 5.txt	50	5	50	Solution	30.02	474	442.00	6.75
436 50 5 50 rep 6.txt	50	5	50	Solution	30.02	483	461.00	4.55
437 50 5 50 rep 7.txt	50	5	50	Solution	30.03	455	428.00	5.93
438 50 5 50 rep 8.txt	50	5	50	Solution	30.02	463	445.00	3.89
439 50 5 50 rep 9.txt	50	5	50	Solution	30.04	502	471.00	6.18
440 50 5 50 rep 10.txt	50	5	50	Solution	30.02	467	432.00	7.49
441 50 10 10 rep 1.txt	50	10	10	Solution	30.05	2384	1878.00	21.22
442 50 10 10 rep 2.txt	50	10	10	Solution	30.08	2231	1742.00	21.92
443 50 10 10 rep 3.txt	50	10	10	Solution	30.04	2340	1963.00	16.11
444 50 10 10 rep 4.txt	50	10	10	Solution	30.03	2649	2164.00	18.31
445 50 10 10 rep 5.txt	50	10	10	Solution	30.04	2206	1791.00	18.81
446 50 10 10 rep 6.txt	50	10	10	Solution	30.04	2370	2096.00	11.56
447 50 10 10 rep 7.txt	50	10	10	Solution	30.07	2352	1880.00	20.07

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
448 50 10 10 rep 8.txt	50	10	10	Solution	30.06	2339	1960.00	16.20
449 50 10 10 rep 9.txt	50	10	10	Solution	30.07	2262	1816.00	19.72
450 50 10 10 rep 10.txt	50	10	10	Solution	30.05	2311	1799.00	22.15
451 50 10 20 rep 1.txt	50	10	20	Solution	30.03	1284	1118.00	12.93
452 50 10 20 rep 2.txt	50	10	20	Solution	30.04	1269	1111.00	12.45
453 50 10 20 rep 3.txt	50	10	20	Solution	30.03	1253	1028.00	17.96
454 50 10 20 rep 4.txt	50	10	20	Solution	30.04	1254	1176.00	6.22
455 50 10 20 rep 5.txt	50	10	20	Solution	30.03	1260	1044.00	17.14
456 50 10 20 rep 6.txt	50	10	20	Solution	30.05	1208	1047.00	13.33
457 50 10 20 rep 7.txt	50	10	20	Solution	30.03	1279	1138.00	11.02
458 50 10 20 rep 8.txt	50	10	20	Solution	30.04	1222	1085.00	11.21
459 50 10 20 rep 9.txt	50	10	20	Solution	30.05	1256	1107.00	11.86
460 50 10 20 rep 10.txt	50	10	20	Solution	30.05	1305	1184.00	9.27
461 50 10 30 rep 1.txt	50	10	30	Solution	30.01	997	862.00	13.54
462 50 10 30 rep 2.txt	50	10	30	Solution	30.03	920	879.00	4.46
463 50 10 30 rep 3.txt	50	10	30	Solution	30.02	926	787.00	15.01
464 50 10 30 rep 4.txt	50	10	30	Solution	30.02	887	788.00	11.16
465 50 10 30 rep 5.txt	50	10	30	Solution	30.04	973	887.00	8.84
466 50 10 30 rep 6.txt	50	10	30	Solution	30.02	954	874.00	8.39
467 50 10 30 rep 7.txt	50	10	30	Solution	30.05	993	871.00	12.29
468 50 10 30 rep 8.txt	50	10	30	Solution	30.03	936	825.00	11.86
469 50 10 30 rep 9.txt	50	10	30	Solution	30.05	977	851.00	12.90
470 50 10 30 rep 10.txt	50	10	30	Solution	30.03	971	910.00	6.28
471 50 10 50 rep 1.txt	50	10	50	Optimal	2.07	727	727.00	0.00
472 50 10 50 rep 2.txt	50	10	50	Solution	30.04	697	695.00	0.29
473 50 10 50 rep 3.txt	50	10	50	Solution	30.03	685	654.00	4.53
474 50 10 50 rep 4.txt	50	10	50	Solution	30.03	747	720.00	3.61
475 50 10 50 rep 5.txt	50	10	50	Solution	30.03	705	696.00	1.28
476 50 10 50 rep 6.txt	50	10	50	Solution	30.06	757	751.00	0.79
477 50 10 50 rep 7.txt	50	10	50	Solution	30.04	756	706.00	6.61
478 50 10 50 rep 8.txt	50	10	50	Optimal	24.71	714	714.00	0.00
479 50 10 50 rep 9.txt	50	10	50	Optimal	1.15	767	767.00	0.00
480 50 10 50 rep 10.txt	50	10	50	Solution	30.02	686	661.00	3.64
481 100 2 10 rep 1.txt	100	2	10	Solution	30.04	2547	2485.00	2.43
482 100 2 10 rep 2.txt	100	2	10	Solution	30.05	3088	2890.00	6.41
483 100 2 10 rep 3.txt	100	2	10	Solution	30.04	2776	2712.00	2.31
484 100 2 10 rep 4.txt	100	2	10	Solution	30.03	2810	2747.00	2.24
485 100 2 10 rep 5.txt	100	2	10	Solution	30.05	2945	2877.00	2.31
486 100 2 10 rep 6.txt	100	2	10	Solution	30.04	3263	3139.00	3.80
487 100 2 10 rep 7.txt	100	2	10	Solution	30.03	2954	2838.00	3.93
488 100 2 10 rep 8.txt	100	2	10	Solution	30.01	3093	2942.00	4.88
489 100 2 10 rep 9.txt	100	2	10	Solution	30.04	2928	2826.00	3.48
490 100 2 10 rep 10.txt	100	2	10	Solution	30.04	2935	2886.00	1.67
491 100 2 20 rep 1.txt	100	2	20	Solution	30.03	1426	1401.00	1.75
492 100 2 20 rep 2.txt	100	2	20	Solution	30.03	1516	1496.00	1.32

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
493 100 2 20 rep 3.txt	100	2	20	Solution	30.01	1325	1289.00	2.72
494 100 2 20 rep 4.txt	100	2	20	Solution	30.03	1325	1284.00	3.09
495 100 2 20 rep 5.txt	100	2	20	Solution	30.02	1398	1378.00	1.43
496 100 2 20 rep 6.txt	100	2	20	Solution	30.02	1459	1439.00	1.37
497 100 2 20 rep 7.txt	100	2	20	Solution	30.04	1431	1405.00	1.82
498 100 2 20 rep 8.txt	100	2	20	Solution	30.03	1507	1481.00	1.73
499 100 2 20 rep 9.txt	100	2	20	Solution	30.02	1337	1320.00	1.27
500 100 2 20 rep 10.txt	100	2	20	Solution	30.03	1451	1419.00	2.21
501 100 2 30 rep 1.txt	100	2	30	Solution	30.01	933	916.00	1.82
502 100 2 30 rep 2.txt	100	2	30	Solution	30.02	873	857.00	1.83
503 100 2 30 rep 3.txt	100	2	30	Solution	30.01	1028	1012.00	1.56
504 100 2 30 rep 4.txt	100	2	30	Solution	30.03	996	980.00	1.61
505 100 2 30 rep 5.txt	100	2	30	Solution	30.03	940	931.00	0.96
506 100 2 30 rep 6.txt	100	2	30	Solution	30.02	920	908.00	1.30
507 100 2 30 rep 7.txt	100	2	30	Solution	30.03	970	956.00	1.44
508 100 2 30 rep 8.txt	100	2	30	Solution	30.02	900	881.00	2.11
509 100 2 30 rep 9.txt	100	2	30	Solution	30.02	905	890.00	1.66
510 100 2 30 rep 10.txt	100	2	30	Solution	30.03	1005	994.00	1.09
511 100 2 50 rep 1.txt	100	2	50	Solution	30.01	658	644.00	2.13
512 100 2 50 rep 2.txt	100	2	50	Solution	30.03	570	558.00	2.11
513 100 2 50 rep 3.txt	100	2	50	Solution	30.02	573	565.00	1.40
514 100 2 50 rep 4.txt	100	2	50	Solution	30.03	614	607.00	1.14
515 100 2 50 rep 5.txt	100	2	50	Solution	30.03	576	567.00	1.56
516 100 2 50 rep 6.txt	100	2	50	Solution	30.04	693	681.00	1.73
517 100 2 50 rep 7.txt	100	2	50	Solution	30.03	587	577.00	1.70
518 100 2 50 rep 8.txt	100	2	50	Solution	30.02	624	614.00	1.60
519 100 2 50 rep 9.txt	100	2	50	Solution	30.04	591	581.00	1.69
520 100 2 50 rep 10.txt	100	2	50	Solution	30.03	547	537.00	1.83
521 100 3 10 rep 1.txt	100	3	10	Solution	30.04	2899	2719.00	6.21
522 100 3 10 rep 2.txt	100	3	10	Solution	30.03	3095	2932.00	5.27
523 100 3 10 rep 3.txt	100	3	10	Solution	30.07	3539	3404.00	3.81
524 100 3 10 rep 4.txt	100	3	10	Solution	30.05	2913	2680.00	8.00
525 100 3 10 rep 5.txt	100	3	10	Solution	30.02	3034	2824.00	6.92
526 100 3 10 rep 6.txt	100	3	10	Solution	30.04	3358	3164.00	5.78
527 100 3 10 rep 7.txt	100	3	10	Solution	30.05	3120	2925.00	6.25
528 100 3 10 rep 8.txt	100	3	10	Solution	30.08	3219	3046.00	5.37
529 100 3 10 rep 9.txt	100	3	10	Solution	30.05	3309	3044.00	8.01
530 100 3 10 rep 10.txt	100	3	10	Solution	30.05	3173	2993.00	5.67
531 100 3 20 rep 1.txt	100	3	20	Solution	30.02	1551	1490.00	3.93
532 100 3 20 rep 2.txt	100	3	20	Solution	30.04	1584	1517.00	4.23
533 100 3 20 rep 3.txt	100	3	20	Solution	30.04	1449	1403.00	3.17
534 100 3 20 rep 4.txt	100	3	20	Solution	30.03	1558	1513.00	2.89
535 100 3 20 rep 5.txt	100	3	20	Solution	30.06	1541	1503.00	2.47
536 100 3 20 rep 6.txt	100	3	20	Solution	30.03	1582	1551.00	1.96
537 100 3 20 rep 7.txt	100	3	20	Solution	30.03	1601	1542.00	3.69

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
538 100 3 20 rep 8.txt	100	3	20	Solution	30.01	1594	1535.00	3.70
539 100 3 20 rep 9.txt	100	3	20	Solution	30.03	1401	1361.00	2.86
540 100 3 20 rep 10.txt	100	3	20	Solution	30.03	1727	1675.00	3.01
541 100 3 30 rep 1.txt	100	3	30	Solution	30.02	1003	978.00	2.49
542 100 3 30 rep 2.txt	100	3	30	Solution	30.03	1028	992.00	3.50
543 100 3 30 rep 3.txt	100	3	30	Solution	30.03	1075	1048.00	2.51
544 100 3 30 rep 4.txt	100	3	30	Solution	30.03	1037	1010.00	2.60
545 100 3 30 rep 5.txt	100	3	30	Solution	30.03	1111	1071.00	3.60
546 100 3 30 rep 6.txt	100	3	30	Solution	30.03	981	946.00	3.57
547 100 3 30 rep 7.txt	100	3	30	Solution	30.04	1048	1020.00	2.67
548 100 3 30 rep 8.txt	100	3	30	Solution	30.05	1002	969.00	3.29
549 100 3 30 rep 9.txt	100	3	30	Solution	30.03	938	903.00	3.73
550 100 3 30 rep 10.txt	100	3	30	Solution	30.01	1132	1099.00	2.92
551 100 3 50 rep 1.txt	100	3	50	Solution	30.01	594	576.00	3.03
552 100 3 50 rep 2.txt	100	3	50	Solution	30.02	687	668.00	2.77
553 100 3 50 rep 3.txt	100	3	50	Solution	30.03	638	625.00	2.04
554 100 3 50 rep 4.txt	100	3	50	Solution	30.03	640	623.00	2.66
555 100 3 50 rep 5.txt	100	3	50	Solution	30.02	630	597.00	5.24
556 100 3 50 rep 6.txt	100	3	50	Solution	30.02	634	607.00	4.26
557 100 3 50 rep 7.txt	100	3	50	Solution	30.05	707	694.00	1.84
558 100 3 50 rep 8.txt	100	3	50	Solution	30.02	633	614.00	3.00
559 100 3 50 rep 9.txt	100	3	50	Solution	30.03	597	582.00	2.51
560 100 3 50 rep 10.txt	100	3	50	Solution	30.03	624	607.00	2.72
561 100 5 10 rep 1.txt	100	5	10	Solution	30.06	3245	2913.00	10.23
562 100 5 10 rep 2.txt	100	5	10	Solution	30.06	3330	2987.00	10.30
563 100 5 10 rep 3.txt	100	5	10	Solution	30.08	3662	3112.00	15.02
564 100 5 10 rep 4.txt	100	5	10	Solution	30.08	3952	3470.00	12.20
565 100 5 10 rep 5.txt	100	5	10	Solution	30.09	3355	3068.00	8.55
566 100 5 10 rep 6.txt	100	5	10	Solution	30.07	3400	3131.00	7.91
567 100 5 10 rep 7.txt	100	5	10	Solution	30.08	3765	3306.00	12.19
568 100 5 10 rep 8.txt	100	5	10	Solution	30.11	3789	3358.00	11.38
569 100 5 10 rep 9.txt	100	5	10	Solution	30.08	3405	3112.00	8.60
570 100 5 10 rep 10.txt	100	5	10	Solution	30.08	3414	3005.00	11.98
571 100 5 20 rep 1.txt	100	5	20	Solution	30.04	1640	1542.00	5.98
572 100 5 20 rep 2.txt	100	5	20	Solution	30.06	1771	1690.00	4.57
573 100 5 20 rep 3.txt	100	5	20	Solution	30.06	1734	1650.00	4.84
574 100 5 20 rep 4.txt	100	5	20	Solution	30.05	1666	1515.00	9.06
575 100 5 20 rep 5.txt	100	5	20	Solution	30.05	1717	1659.00	3.38
576 100 5 20 rep 6.txt	100	5	20	Solution	30.04	1714	1601.00	6.59
577 100 5 20 rep 7.txt	100	5	20	Solution	30.05	1652	1574.00	4.72
578 100 5 20 rep 8.txt	100	5	20	Solution	30.05	1762	1602.00	9.08
579 100 5 20 rep 9.txt	100	5	20	Solution	30.04	1597	1509.00	5.51
580 100 5 20 rep 10.txt	100	5	20	Solution	30.06	1675	1599.00	4.54
581 100 5 30 rep 1.txt	100	5	30	Solution	30.05	1071	989.00	7.66
582 100 5 30 rep 2.txt	100	5	30	Solution	30.04	1130	1001.00	11.42

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
583 100 5 30 rep 3.txt	100	5	30	Solution	30.06	1121	994.00	11.33
584 100 5 30 rep 4.txt	100	5	30	Solution	30.04	1155	1113.00	3.64
585 100 5 30 rep 5.txt	100	5	30	Solution	30.04	1130	1046.00	7.43
586 100 5 30 rep 6.txt	100	5	30	Solution	30.03	1143	1067.00	6.65
587 100 5 30 rep 7.txt	100	5	30	Solution	30.01	1167	1086.00	6.94
588 100 5 30 rep 8.txt	100	5	30	Solution	30.03	1148	1063.00	7.40
589 100 5 30 rep 9.txt	100	5	30	Solution	30.04	1157	1095.00	5.36
590 100 5 30 rep 10.txt	100	5	30	Solution	30.04	1225	1166.00	4.82
591 100 5 50 rep 1.txt	100	5	50	Solution	30.04	772	742.00	3.89
592 100 5 50 rep 2.txt	100	5	50	Solution	30.03	710	655.00	7.75
593 100 5 50 rep 3.txt	100	5	50	Solution	30.05	780	724.00	7.18
594 100 5 50 rep 4.txt	100	5	50	Solution	30.04	760	699.00	8.03
595 100 5 50 rep 5.txt	100	5	50	Solution	30.04	749	693.00	7.48
596 100 5 50 rep 6.txt	100	5	50	Solution	30.03	785	765.00	2.55
597 100 5 50 rep 7.txt	100	5	50	Solution	30.06	726	699.00	3.72
598 100 5 50 rep 8.txt	100	5	50	Solution	30.03	756	719.00	4.89
599 100 5 50 rep 9.txt	100	5	50	Solution	30.05	746	712.00	4.56
600 100 5 50 rep 10.txt	100	5	50	Solution	30.03	719	698.00	2.92
601 100 10 10 rep 1.txt	100	10	10	Solution	30.16	4101	3440.00	16.12
602 100 10 10 rep 2.txt	100	10	10	Solution	30.21	4346	3511.00	19.21
603 100 10 10 rep 3.txt	100	10	10	Solution	30.13	3946	3186.00	19.26
604 100 10 10 rep 4.txt	100	10	10	Solution	30.20	4232	3469.00	18.03
605 100 10 10 rep 5.txt	100	10	10	Solution	30.14	4379	3629.00	17.13
606 100 10 10 rep 6.txt	100	10	10	Solution	30.16	4229	3483.00	17.64
607 100 10 10 rep 7.txt	100	10	10	Solution	30.15	4221	3418.00	19.02
608 100 10 10 rep 8.txt	100	10	10	Solution	30.12	4065	3173.00	21.94
609 100 10 10 rep 9.txt	100	10	10	Solution	30.18	4117	3261.00	20.79
610 100 10 10 rep 10.txt	100	10	10	Solution	30.15	4215	3380.00	19.81
611 100 10 20 rep 1.txt	100	10	20	Solution	30.12	2099	1748.00	16.72
612 100 10 20 rep 2.txt	100	10	20	Solution	30.11	2107	1740.00	17.42
613 100 10 20 rep 3.txt	100	10	20	Solution	30.06	2162	1740.00	19.52
614 100 10 20 rep 4.txt	100	10	20	Solution	30.11	2126	1818.00	14.49
615 100 10 20 rep 5.txt	100	10	20	Solution	30.17	2156	1903.00	11.73
616 100 10 20 rep 6.txt	100	10	20	Solution	30.11	2186	1883.00	13.86
617 100 10 20 rep 7.txt	100	10	20	Solution	30.10	2108	1725.00	18.17
618 100 10 20 rep 8.txt	100	10	20	Solution	30.13	2163	1766.00	18.35
619 100 10 20 rep 9.txt	100	10	20	Solution	30.13	2062	1690.00	18.04
620 100 10 20 rep 10.txt	100	10	20	Solution	30.06	2017	1794.00	11.06
621 100 10 30 rep 1.txt	100	10	30	Solution	30.11	1517	1338.00	11.80
622 100 10 30 rep 2.txt	100	10	30	Solution	30.09	1474	1279.00	13.23
623 100 10 30 rep 3.txt	100	10	30	Solution	30.09	1443	1219.00	15.52
624 100 10 30 rep 4.txt	100	10	30	Solution	30.14	1483	1256.00	15.31
625 100 10 30 rep 5.txt	100	10	30	Solution	30.09	1494	1371.00	8.23
626 100 10 30 rep 6.txt	100	10	30	Solution	30.11	1550	1315.00	15.16
627 100 10 30 rep 7.txt	100	10	30	Solution	30.05	1451	1271.00	12.41

Table 1: Results for CFS (CPO) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
628 100 10 30 rep 8.txt	100	10	30	Solution	30.10	1500	1324.00	11.73
629 100 10 30 rep 9.txt	100	10	30	Solution	30.11	1489	1259.00	15.45
630 100 10 30 rep 10.txt	100	10	30	Solution	30.07	1518	1300.00	14.36
631 100 10 50 rep 1.txt	100	10	50	Solution	30.11	1068	995.00	6.84
632 100 10 50 rep 2.txt	100	10	50	Solution	30.06	1035	886.00	14.40
633 100 10 50 rep 3.txt	100	10	50	Solution	30.09	1027	916.00	10.81
634 100 10 50 rep 4.txt	100	10	50	Solution	30.14	996	847.00	14.96
635 100 10 50 rep 5.txt	100	10	50	Solution	30.07	987	870.00	11.85
636 100 10 50 rep 6.txt	100	10	50	Solution	30.11	1037	927.00	10.61
637 100 10 50 rep 7.txt	100	10	50	Solution	30.12	1004	873.00	13.05
638 100 10 50 rep 8.txt	100	10	50	Solution	30.10	990	910.00	8.08
639 100 10 50 rep 9.txt	100	10	50	Solution	30.12	1027	904.00	11.98
640 100 10 50 rep 10.txt	100	10	50	Solution	30.14	1079	974.00	9.73

2 Instances (CPSat)

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
1 20 2 10 rep 1.txt	20	2	10	Solution	300.19	721	695.00	3.61
2 20 2 10 rep 2.txt	20	2	10	Solution	300.20	610	607.00	0.49
3 20 2 10 rep 3.txt	20	2	10	Optimal	300.04	553	553.00	0.00
4 20 2 10 rep 4.txt	20	2	10	Optimal	300.03	794	794.00	0.00
5 20 2 10 rep 5.txt	20	2	10	Solution	300.14	561	542.00	3.39
6 20 2 10 rep 6.txt	20	2	10	Solution	300.94	648	639.00	1.39
7 20 2 10 rep 7.txt	20	2	10	Solution	300.13	620	612.00	1.29
8 20 2 10 rep 8.txt	20	2	10	Optimal	300.03	946	946.00	0.00
9 20 2 10 rep 9.txt	20	2	10	Optimal	36.12	548	548.00	0.00
10 20 2 10 rep 10.txt	20	2	10	Optimal	12.54	745	745.00	0.00
11 20 2 20 rep 1.txt	20	2	20	Optimal	245.63	323	323.00	0.00
12 20 2 20 rep 2.txt	20	2	20	Optimal	300.03	320	320.00	0.00
13 20 2 20 rep 3.txt	20	2	20	Solution	300.11	289	286.00	1.04
14 20 2 20 rep 4.txt	20	2	20	Optimal	300.01	299	299.00	0.00
15 20 2 20 rep 5.txt	20	2	20	Optimal	48.26	320	320.00	0.00
16 20 2 20 rep 6.txt	20	2	20	Optimal	300.03	327	327.00	0.00
17 20 2 20 rep 7.txt	20	2	20	Optimal	300.02	352	352.00	0.00
18 20 2 20 rep 8.txt	20	2	20	Optimal	300.02	344	344.00	0.00
19 20 2 20 rep 9.txt	20	2	20	Optimal	300.03	290	290.00	0.00
20 20 2 20 rep 10.txt	20	2	20	Solution	300.13	355	350.00	1.41
21 20 2 30 rep 1.txt	20	2	30	Optimal	82.81	185	185.00	0.00
22 20 2 30 rep 2.txt	20	2	30	Optimal	0.31	164	164.00	0.00

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
23 20 2 30 rep 3.txt	20	2	30	Optimal	17.99	231	231.00	0.00
24 20 2 30 rep 4.txt	20	2	30	Optimal	300.03	255	255.00	0.00
25 20 2 30 rep 5.txt	20	2	30	Optimal	300.02	235	235.00	0.00
26 20 2 30 rep 6.txt	20	2	30	Optimal	13.96	253	253.00	0.00
27 20 2 30 rep 7.txt	20	2	30	Optimal	300.03	209	209.00	0.00
28 20 2 30 rep 8.txt	20	2	30	Optimal	0.30	228	228.00	0.00
29 20 2 30 rep 9.txt	20	2	30	Optimal	300.02	234	234.00	0.00
30 20 2 30 rep 10.txt	20	2	30	Optimal	300.03	337	337.00	0.00
31 20 2 50 rep 1.txt	20	2	50	Optimal	0.03	191	191.00	0.00
32 20 2 50 rep 2.txt	20	2	50	Optimal	0.04	172	172.00	0.00
33 20 2 50 rep 3.txt	20	2	50	Optimal	0.20	173	173.00	0.00
34 20 2 50 rep 4.txt	20	2	50	Optimal	0.03	184	184.00	0.00
35 20 2 50 rep 5.txt	20	2	50	Optimal	0.19	179	179.00	0.00
36 20 2 50 rep 6.txt	20	2	50	Optimal	300.02	220	220.00	0.00
37 20 2 50 rep 7.txt	20	2	50	Optimal	0.04	170	170.00	0.00
38 20 2 50 rep 8.txt	20	2	50	Optimal	0.03	192	192.00	0.00
39 20 2 50 rep 9.txt	20	2	50	Optimal	1.36	204	204.00	0.00
40 20 2 50 rep 10.txt	20	2	50	Optimal	0.36	217	217.00	0.00
41 20 3 10 rep 1.txt	20	3	10	Solution	300.12	726	705.00	2.89
42 20 3 10 rep 2.txt	20	3	10	Optimal	300.03	719	719.00	0.00
43 20 3 10 rep 3.txt	20	3	10	Optimal	300.01	657	657.00	0.00
44 20 3 10 rep 4.txt	20	3	10	Optimal	0.09	1104	1104.00	0.00
45 20 3 10 rep 5.txt	20	3	10	Optimal	300.03	725	725.00	0.00
46 20 3 10 rep 6.txt	20	3	10	Optimal	59.53	888	888.00	0.00
47 20 3 10 rep 7.txt	20	3	10	Solution	300.17	832	824.00	0.96
48 20 3 10 rep 8.txt	20	3	10	Solution	300.09	735	725.00	1.36
49 20 3 10 rep 9.txt	20	3	10	Optimal	11.58	876	876.00	0.00
50 20 3 10 rep 10.txt	20	3	10	Optimal	300.03	923	923.00	0.00
51 20 3 20 rep 1.txt	20	3	20	Solution	300.20	448	441.00	1.56
52 20 3 20 rep 2.txt	20	3	20	Solution	300.10	381	372.00	2.36
53 20 3 20 rep 3.txt	20	3	20	Optimal	300.03	385	385.00	0.00
54 20 3 20 rep 4.txt	20	3	20	Optimal	21.87	380	380.00	0.00
55 20 3 20 rep 5.txt	20	3	20	Optimal	300.02	394	394.00	0.00
56 20 3 20 rep 6.txt	20	3	20	Optimal	11.18	350	350.00	0.00
57 20 3 20 rep 7.txt	20	3	20	Optimal	18.55	431	431.00	0.00
58 20 3 20 rep 8.txt	20	3	20	Optimal	300.03	331	331.00	0.00
59 20 3 20 rep 9.txt	20	3	20	Solution	300.13	358	348.00	2.79
60 20 3 20 rep 10.txt	20	3	20	Optimal	300.02	365	365.00	0.00
61 20 3 30 rep 1.txt	20	3	30	Optimal	300.01	338	338.00	0.00
62 20 3 30 rep 2.txt	20	3	30	Optimal	300.03	241	241.00	0.00
63 20 3 30 rep 3.txt	20	3	30	Optimal	5.82	269	269.00	0.00
64 20 3 30 rep 4.txt	20	3	30	Optimal	300.02	304	304.00	0.00
65 20 3 30 rep 5.txt	20	3	30	Optimal	24.99	263	263.00	0.00
66 20 3 30 rep 6.txt	20	3	30	Optimal	300.02	259	259.00	0.00
67 20 3 30 rep 7.txt	20	3	30	Optimal	300.02	377	377.00	0.00

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
68 20 3 30 rep 8.txt	20	3	30	Optimal	0.90	287	287.00	0.00
69 20 3 30 rep 9.txt	20	3	30	Optimal	300.02	241	241.00	0.00
70 20 3 30 rep 10.txt	20	3	30	Optimal	1.49	228	228.00	0.00
71 20 3 50 rep 1.txt	20	3	50	Optimal	0.02	259	259.00	0.00
72 20 3 50 rep 2.txt	20	3	50	Optimal	0.03	233	233.00	0.00
73 20 3 50 rep 3.txt	20	3	50	Optimal	0.02	234	234.00	0.00
74 20 3 50 rep 4.txt	20	3	50	Optimal	0.04	270	270.00	0.00
75 20 3 50 rep 5.txt	20	3	50	Optimal	0.03	247	247.00	0.00
76 20 3 50 rep 6.txt	20	3	50	Optimal	0.22	245	245.00	0.00
77 20 3 50 rep 7.txt	20	3	50	Optimal	211.34	217	217.00	0.00
78 20 3 50 rep 8.txt	20	3	50	Optimal	0.04	232	232.00	0.00
79 20 3 50 rep 9.txt	20	3	50	Optimal	8.89	241	241.00	0.00
80 20 3 50 rep 10.txt	20	3	50	Optimal	0.04	223	223.00	0.00
81 20 5 10 rep 1.txt	20	5	10	Solution	300.08	708	651.00	8.05
82 20 5 10 rep 2.txt	20	5	10	Solution	300.12	926	915.00	1.19
83 20 5 10 rep 3.txt	20	5	10	Solution	300.15	829	811.00	2.17
84 20 5 10 rep 4.txt	20	5	10	Solution	300.40	767	751.00	2.09
85 20 5 10 rep 5.txt	20	5	10	Optimal	300.03	774	774.00	0.00
86 20 5 10 rep 6.txt	20	5	10	Optimal	300.01	995	995.00	0.00
87 20 5 10 rep 7.txt	20	5	10	Solution	300.44	743	702.00	5.52
88 20 5 10 rep 8.txt	20	5	10	Solution	300.77	917	900.00	1.85
89 20 5 10 rep 9.txt	20	5	10	Optimal	4.77	881	881.00	0.00
90 20 5 10 rep 10.txt	20	5	10	Solution	300.12	931	913.00	1.93
91 20 5 20 rep 1.txt	20	5	20	Solution	300.71	481	454.00	5.61
92 20 5 20 rep 2.txt	20	5	20	Solution	300.08	498	466.00	6.43
93 20 5 20 rep 3.txt	20	5	20	Solution	300.81	510	489.00	4.12
94 20 5 20 rep 4.txt	20	5	20	Optimal	300.03	447	447.00	0.00
95 20 5 20 rep 5.txt	20	5	20	Solution	300.17	546	509.00	6.78
96 20 5 20 rep 6.txt	20	5	20	Solution	300.10	489	473.00	3.27
97 20 5 20 rep 7.txt	20	5	20	Solution	300.45	550	534.00	2.91
98 20 5 20 rep 8.txt	20	5	20	Solution	300.12	475	452.00	4.84
99 20 5 20 rep 9.txt	20	5	20	Optimal	300.01	451	451.00	0.00
100 20 5 20 rep 10.txt	20	5	20	Optimal	300.03	493	493.00	0.00
101 20 5 30 rep 1.txt	20	5	30	Optimal	2.92	370	370.00	0.00
102 20 5 30 rep 2.txt	20	5	30	Solution	300.07	386	378.00	2.07
103 20 5 30 rep 3.txt	20	5	30	Optimal	91.09	371	371.00	0.00
104 20 5 30 rep 4.txt	20	5	30	Optimal	300.02	400	400.00	0.00
105 20 5 30 rep 5.txt	20	5	30	Optimal	300.03	405	405.00	0.00
106 20 5 30 rep 6.txt	20	5	30	Optimal	300.02	388	388.00	0.00
107 20 5 30 rep 7.txt	20	5	30	Optimal	0.52	424	424.00	0.00
108 20 5 30 rep 8.txt	20	5	30	Optimal	0.07	407	407.00	0.00
109 20 5 30 rep 9.txt	20	5	30	Optimal	300.02	358	358.00	0.00
110 20 5 30 rep 10.txt	20	5	30	Optimal	2.64	427	427.00	0.00
111 20 5 50 rep 1.txt	20	5	50	Optimal	0.03	408	408.00	0.00
112 20 5 50 rep 2.txt	20	5	50	Optimal	0.03	418	418.00	0.00

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
113 20 5 50 rep 3.txt	20	5	50	Optimal	0.03	357	357.00	0.00
114 20 5 50 rep 4.txt	20	5	50	Optimal	0.05	428	428.00	0.00
115 20 5 50 rep 5.txt	20	5	50	Optimal	0.06	373	373.00	0.00
116 20 5 50 rep 6.txt	20	5	50	Optimal	0.04	365	365.00	0.00
117 20 5 50 rep 7.txt	20	5	50	Optimal	0.04	387	387.00	0.00
118 20 5 50 rep 8.txt	20	5	50	Optimal	0.04	348	348.00	0.00
119 20 5 50 rep 9.txt	20	5	50	Optimal	0.03	405	405.00	0.00
120 20 5 50 rep 10.txt	20	5	50	Optimal	0.02	417	417.00	0.00
121 20 10 10 rep 1.txt	20	10	10	Solution	300.08	1120	1036.00	7.50
122 20 10 10 rep 2.txt	20	10	10	Solution	300.11	1252	1110.00	11.34
123 20 10 10 rep 3.txt	20	10	10	Solution	300.06	1216	1158.00	4.77
124 20 10 10 rep 4.txt	20	10	10	Solution	300.53	1184	1107.00	6.50
125 20 10 10 rep 5.txt	20	10	10	Solution	300.09	1206	1137.00	5.72
126 20 10 10 rep 6.txt	20	10	10	Solution	300.11	1157	1067.00	7.78
127 20 10 10 rep 7.txt	20	10	10	Solution	300.10	1149	1119.00	2.61
128 20 10 10 rep 8.txt	20	10	10	Solution	300.16	1374	1255.00	8.66
129 20 10 10 rep 9.txt	20	10	10	Solution	300.04	1165	1144.00	1.80
130 20 10 10 rep 10.txt	20	10	10	Solution	300.07	1257	1198.00	4.69
131 20 10 20 rep 1.txt	20	10	20	Optimal	300.01	734	734.00	0.00
132 20 10 20 rep 2.txt	20	10	20	Solution	300.04	773	742.00	4.01
133 20 10 20 rep 3.txt	20	10	20	Solution	300.05	749	707.00	5.61
134 20 10 20 rep 4.txt	20	10	20	Solution	300.08	754	711.00	5.70
135 20 10 20 rep 5.txt	20	10	20	Solution	300.07	841	779.00	7.37
136 20 10 20 rep 6.txt	20	10	20	Solution	300.07	813	762.00	6.27
137 20 10 20 rep 7.txt	20	10	20	Solution	300.29	808	763.00	5.57
138 20 10 20 rep 8.txt	20	10	20	Solution	300.25	747	687.00	8.03
139 20 10 20 rep 9.txt	20	10	20	Solution	300.09	816	757.00	7.23
140 20 10 20 rep 10.txt	20	10	20	Solution	300.38	767	745.00	2.87
141 20 10 30 rep 1.txt	20	10	30	Optimal	300.01	713	713.00	0.00
142 20 10 30 rep 2.txt	20	10	30	Optimal	0.17	720	720.00	0.00
143 20 10 30 rep 3.txt	20	10	30	Optimal	6.60	682	682.00	0.00
144 20 10 30 rep 4.txt	20	10	30	Optimal	7.56	642	642.00	0.00
145 20 10 30 rep 5.txt	20	10	30	Optimal	37.11	697	697.00	0.00
146 20 10 30 rep 6.txt	20	10	30	Optimal	300.01	616	616.00	0.00
147 20 10 30 rep 7.txt	20	10	30	Optimal	300.02	643	643.00	0.00
148 20 10 30 rep 8.txt	20	10	30	Optimal	0.10	631	631.00	0.00
149 20 10 30 rep 9.txt	20	10	30	Optimal	114.80	680	680.00	0.00
150 20 10 30 rep 10.txt	20	10	30	Optimal	300.00	692	692.00	0.00
151 20 10 50 rep 1.txt	20	10	50	Optimal	0.03	711	711.00	0.00
152 20 10 50 rep 2.txt	20	10	50	Optimal	0.02	621	621.00	0.00
153 20 10 50 rep 3.txt	20	10	50	Optimal	0.02	669	669.00	0.00
154 20 10 50 rep 4.txt	20	10	50	Optimal	0.03	740	740.00	0.00
155 20 10 50 rep 5.txt	20	10	50	Optimal	0.03	686	686.00	0.00
156 20 10 50 rep 6.txt	20	10	50	Optimal	0.02	715	715.00	0.00
157 20 10 50 rep 7.txt	20	10	50	Optimal	0.02	612	612.00	0.00

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
158 20 10 50 rep 8.txt	20	10	50	Optimal	0.03	628	628.00	0.00
159 20 10 50 rep 9.txt	20	10	50	Optimal	0.04	744	744.00	0.00
160 20 10 50 rep 10.txt	20	10	50	Optimal	0.02	707	707.00	0.00
161 30 2 10 rep 1.txt	30	2	10	Solution	300.23	1255	1249.00	0.48
162 30 2 10 rep 2.txt	30	2	10	Solution	300.13	826	811.00	1.82
163 30 2 10 rep 3.txt	30	2	10	Solution	300.89	852	842.00	1.17
164 30 2 10 rep 4.txt	30	2	10	Solution	300.20	1078	1074.00	0.37
165 30 2 10 rep 5.txt	30	2	10	Optimal	1.76	1177	1177.00	0.00
166 30 2 10 rep 6.txt	30	2	10	Solution	300.19	846	832.00	1.65
167 30 2 10 rep 7.txt	30	2	10	Solution	300.12	712	705.00	0.98
168 30 2 10 rep 8.txt	30	2	10	Optimal	300.04	890	890.00	0.00
169 30 2 10 rep 9.txt	30	2	10	Optimal	300.02	983	983.00	0.00
170 30 2 10 rep 10.txt	30	2	10	Solution	300.12	1140	1139.00	0.09
171 30 2 20 rep 1.txt	30	2	20	Solution	301.21	420	416.00	0.95
172 30 2 20 rep 2.txt	30	2	20	Solution	300.15	514	506.00	1.56
173 30 2 20 rep 3.txt	30	2	20	Solution	300.08	403	398.00	1.24
174 30 2 20 rep 4.txt	30	2	20	Solution	300.12	422	418.00	0.95
175 30 2 20 rep 5.txt	30	2	20	Solution	300.13	449	442.00	1.56
176 30 2 20 rep 6.txt	30	2	20	Solution	300.11	433	427.00	1.39
177 30 2 20 rep 7.txt	30	2	20	Solution	300.26	486	483.00	0.62
178 30 2 20 rep 8.txt	30	2	20	Solution	300.18	389	378.00	2.83
179 30 2 20 rep 9.txt	30	2	20	Solution	300.64	433	428.00	1.15
180 30 2 20 rep 10.txt	30	2	20	Solution	300.30	545	524.00	3.85
181 30 2 30 rep 1.txt	30	2	30	Solution	300.08	312	309.00	0.96
182 30 2 30 rep 2.txt	30	2	30	Optimal	300.02	229	229.00	0.00
183 30 2 30 rep 3.txt	30	2	30	Solution	300.06	331	329.00	0.60
184 30 2 30 rep 4.txt	30	2	30	Solution	300.08	326	324.00	0.61
185 30 2 30 rep 5.txt	30	2	30	Solution	300.11	289	286.00	1.04
186 30 2 30 rep 6.txt	30	2	30	Solution	300.20	337	332.00	1.48
187 30 2 30 rep 7.txt	30	2	30	Solution	300.10	373	368.00	1.34
188 30 2 30 rep 8.txt	30	2	30	Solution	300.06	348	345.00	0.86
189 30 2 30 rep 9.txt	30	2	30	Solution	300.10	396	392.00	1.01
190 30 2 30 rep 10.txt	30	2	30	Solution	300.63	312	307.00	1.60
191 30 2 50 rep 1.txt	30	2	50	Optimal	300.01	203	203.00	0.00
192 30 2 50 rep 2.txt	30	2	50	Optimal	1.00	195	195.00	0.00
193 30 2 50 rep 3.txt	30	2	50	Solution	300.05	205	202.00	1.46
194 30 2 50 rep 4.txt	30	2	50	Optimal	37.55	201	201.00	0.00
195 30 2 50 rep 5.txt	30	2	50	Optimal	300.01	195	195.00	0.00
196 30 2 50 rep 6.txt	30	2	50	Solution	300.04	218	216.00	0.92
197 30 2 50 rep 7.txt	30	2	50	Solution	300.03	242	239.00	1.24
198 30 2 50 rep 8.txt	30	2	50	Optimal	0.02	176	176.00	0.00
199 30 2 50 rep 9.txt	30	2	50	Optimal	0.99	174	174.00	0.00
200 30 2 50 rep 10.txt	30	2	50	Optimal	300.01	229	229.00	0.00
201 30 3 10 rep 1.txt	30	3	10	Optimal	0.12	1069	1069.00	0.00
202 30 3 10 rep 2.txt	30	3	10	Optimal	300.02	1062	1062.00	0.00

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
203 30 3 10 rep 3.txt	30	3	10	Solution	300.14	1165	1148.00	1.46
204 30 3 10 rep 4.txt	30	3	10	Solution	300.14	874	855.00	2.17
205 30 3 10 rep 5.txt	30	3	10	Optimal	300.01	1059	1059.00	0.00
206 30 3 10 rep 6.txt	30	3	10	Solution	300.13	1025	1020.00	0.49
207 30 3 10 rep 7.txt	30	3	10	Solution	300.34	907	880.00	2.98
208 30 3 10 rep 8.txt	30	3	10	Solution	300.60	810	802.00	0.99
209 30 3 10 rep 9.txt	30	3	10	Optimal	300.01	1107	1107.00	0.00
210 30 3 10 rep 10.txt	30	3	10	Solution	300.26	968	956.00	1.24
211 30 3 20 rep 1.txt	30	3	20	Solution	300.13	529	519.00	1.89
212 30 3 20 rep 2.txt	30	3	20	Solution	300.20	595	586.00	1.51
213 30 3 20 rep 3.txt	30	3	20	Solution	300.16	465	451.00	3.01
214 30 3 20 rep 4.txt	30	3	20	Solution	300.13	596	579.00	2.85
215 30 3 20 rep 5.txt	30	3	20	Solution	300.27	535	527.00	1.50
216 30 3 20 rep 6.txt	30	3	20	Solution	300.15	585	573.00	2.05
217 30 3 20 rep 7.txt	30	3	20	Solution	300.17	618	607.00	1.78
218 30 3 20 rep 8.txt	30	3	20	Solution	300.11	555	542.00	2.34
219 30 3 20 rep 9.txt	30	3	20	Solution	300.14	530	515.00	2.83
220 30 3 20 rep 10.txt	30	3	20	Solution	300.20	511	505.00	1.17
221 30 3 30 rep 1.txt	30	3	30	Solution	300.06	371	368.00	0.81
222 30 3 30 rep 2.txt	30	3	30	Solution	300.10	341	326.00	4.40
223 30 3 30 rep 3.txt	30	3	30	Solution	300.07	429	421.00	1.86
224 30 3 30 rep 4.txt	30	3	30	Solution	300.06	380	366.00	3.68
225 30 3 30 rep 5.txt	30	3	30	Solution	300.06	306	302.00	1.31
226 30 3 30 rep 6.txt	30	3	30	Solution	300.06	364	361.00	0.82
227 30 3 30 rep 7.txt	30	3	30	Solution	300.07	391	388.00	0.77
228 30 3 30 rep 8.txt	30	3	30	Solution	300.09	362	350.00	3.31
229 30 3 30 rep 9.txt	30	3	30	Optimal	300.01	349	349.00	0.00
230 30 3 30 rep 10.txt	30	3	30	Solution	300.05	335	318.00	5.07
231 30 3 50 rep 1.txt	30	3	50	Optimal	40.98	259	259.00	0.00
232 30 3 50 rep 2.txt	30	3	50	Optimal	0.45	238	238.00	0.00
233 30 3 50 rep 3.txt	30	3	50	Solution	300.03	259	257.00	0.77
234 30 3 50 rep 4.txt	30	3	50	Optimal	300.01	267	267.00	0.00
235 30 3 50 rep 5.txt	30	3	50	Optimal	0.02	256	256.00	0.00
236 30 3 50 rep 6.txt	30	3	50	Optimal	43.28	260	260.00	0.00
237 30 3 50 rep 7.txt	30	3	50	Optimal	300.01	266	266.00	0.00
238 30 3 50 rep 8.txt	30	3	50	Optimal	67.59	261	261.00	0.00
239 30 3 50 rep 9.txt	30	3	50	Optimal	300.01	236	236.00	0.00
240 30 3 50 rep 10.txt	30	3	50	Optimal	35.37	263	263.00	0.00
241 30 5 10 rep 1.txt	30	5	10	Solution	300.16	1160	1124.00	3.10
242 30 5 10 rep 2.txt	30	5	10	Solution	300.08	1093	1020.00	6.68
243 30 5 10 rep 3.txt	30	5	10	Solution	300.08	1177	1156.00	1.78
244 30 5 10 rep 4.txt	30	5	10	Solution	300.11	1047	1003.00	4.20
245 30 5 10 rep 5.txt	30	5	10	Solution	300.12	1108	1057.00	4.60
246 30 5 10 rep 6.txt	30	5	10	Solution	300.07	1285	1257.00	2.18
247 30 5 10 rep 7.txt	30	5	10	Optimal	300.02	1130	1130.00	0.00

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
248 30 5 10 rep 8.txt	30	5	10	Solution	300.07	958	922.00	3.76
249 30 5 10 rep 9.txt	30	5	10	Solution	300.12	1106	1028.00	7.05
250 30 5 10 rep 10.txt	30	5	10	Solution	300.09	1168	1137.00	2.65
251 30 5 20 rep 1.txt	30	5	20	Solution	300.12	659	634.00	3.79
252 30 5 20 rep 2.txt	30	5	20	Solution	300.15	674	662.00	1.78
253 30 5 20 rep 3.txt	30	5	20	Solution	300.10	624	580.00	7.05
254 30 5 20 rep 4.txt	30	5	20	Solution	300.11	622	618.00	0.64
255 30 5 20 rep 5.txt	30	5	20	Solution	300.10	602	561.00	6.81
256 30 5 20 rep 6.txt	30	5	20	Solution	300.11	585	568.00	2.91
257 30 5 20 rep 7.txt	30	5	20	Solution	300.10	655	612.00	6.56
258 30 5 20 rep 8.txt	30	5	20	Solution	300.12	597	570.00	4.52
259 30 5 20 rep 9.txt	30	5	20	Solution	300.11	673	631.00	6.24
260 30 5 20 rep 10.txt	30	5	20	Solution	300.26	693	656.00	5.34
261 30 5 30 rep 1.txt	30	5	30	Solution	300.06	424	415.00	2.12
262 30 5 30 rep 2.txt	30	5	30	Solution	300.27	453	429.00	5.30
263 30 5 30 rep 3.txt	30	5	30	Solution	300.07	453	444.00	1.99
264 30 5 30 rep 4.txt	30	5	30	Optimal	300.01	441	441.00	0.00
265 30 5 30 rep 5.txt	30	5	30	Solution	300.06	450	429.00	4.67
266 30 5 30 rep 6.txt	30	5	30	Solution	300.05	489	452.00	7.57
267 30 5 30 rep 7.txt	30	5	30	Solution	300.05	467	447.00	4.28
268 30 5 30 rep 8.txt	30	5	30	Solution	300.07	515	491.00	4.66
269 30 5 30 rep 9.txt	30	5	30	Solution	300.10	473	456.00	3.59
270 30 5 30 rep 10.txt	30	5	30	Solution	301.21	497	490.00	1.41
271 30 5 50 rep 1.txt	30	5	50	Solution	300.07	397	376.00	5.29
272 30 5 50 rep 2.txt	30	5	50	Optimal	300.04	367	367.00	0.00
273 30 5 50 rep 3.txt	30	5	50	Optimal	5.09	397	397.00	0.00
274 30 5 50 rep 4.txt	30	5	50	Optimal	300.01	411	411.00	0.00
275 30 5 50 rep 5.txt	30	5	50	Optimal	6.81	399	399.00	0.00
276 30 5 50 rep 6.txt	30	5	50	Optimal	0.20	392	392.00	0.00
277 30 5 50 rep 7.txt	30	5	50	Optimal	0.10	406	406.00	0.00
278 30 5 50 rep 8.txt	30	5	50	Optimal	300.03	400	400.00	0.00
279 30 5 50 rep 9.txt	30	5	50	Optimal	0.09	352	352.00	0.00
280 30 5 50 rep 10.txt	30	5	50	Optimal	73.97	399	399.00	0.00
281 30 10 10 rep 1.txt	30	10	10	Solution	300.15	1560	1367.00	12.37
282 30 10 10 rep 2.txt	30	10	10	Solution	300.08	1663	1408.00	15.33
283 30 10 10 rep 3.txt	30	10	10	Solution	300.17	1384	1332.00	3.76
284 30 10 10 rep 4.txt	30	10	10	Solution	300.13	1418	1279.00	9.80
285 30 10 10 rep 5.txt	30	10	10	Solution	300.07	1538	1383.00	10.08
286 30 10 10 rep 6.txt	30	10	10	Solution	300.06	1568	1385.00	11.67
287 30 10 10 rep 7.txt	30	10	10	Solution	300.06	1515	1432.00	5.48
288 30 10 10 rep 8.txt	30	10	10	Solution	300.08	1471	1367.00	7.07
289 30 10 10 rep 9.txt	30	10	10	Solution	300.07	1561	1539.00	1.41
290 30 10 10 rep 10.txt	30	10	10	Solution	300.06	1519	1351.00	11.06
291 30 10 20 rep 1.txt	30	10	20	Solution	300.06	897	769.00	14.27
292 30 10 20 rep 2.txt	30	10	20	Solution	300.07	971	855.00	11.95

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
293 30 10 20 rep 3.txt	30	10	20	Solution	300.06	897	783.00	12.71
294 30 10 20 rep 4.txt	30	10	20	Solution	300.07	858	742.00	13.52
295 30 10 20 rep 5.txt	30	10	20	Solution	300.06	944	833.00	11.76
296 30 10 20 rep 6.txt	30	10	20	Solution	300.06	923	864.00	6.39
297 30 10 20 rep 7.txt	30	10	20	Solution	300.08	892	796.00	10.76
298 30 10 20 rep 8.txt	30	10	20	Solution	300.06	953	845.00	11.33
299 30 10 20 rep 9.txt	30	10	20	Solution	300.08	890	786.00	11.69
300 30 10 20 rep 10.txt	30	10	20	Solution	300.07	889	801.00	9.90
301 30 10 30 rep 1.txt	30	10	30	Solution	300.06	727	712.00	2.06
302 30 10 30 rep 2.txt	30	10	30	Solution	300.05	758	718.00	5.28
303 30 10 30 rep 3.txt	30	10	30	Solution	300.04	726	687.00	5.37
304 30 10 30 rep 4.txt	30	10	30	Solution	300.05	697	675.00	3.16
305 30 10 30 rep 5.txt	30	10	30	Optimal	300.01	777	777.00	0.00
306 30 10 30 rep 6.txt	30	10	30	Solution	300.11	745	717.00	3.76
307 30 10 30 rep 7.txt	30	10	30	Solution	300.05	732	681.00	6.97
308 30 10 30 rep 8.txt	30	10	30	Solution	300.05	718	674.00	6.13
309 30 10 30 rep 9.txt	30	10	30	Solution	300.08	738	700.00	5.15
310 30 10 30 rep 10.txt	30	10	30	Solution	300.06	708	695.00	1.84
311 30 10 50 rep 1.txt	30	10	50	Optimal	0.10	760	760.00	0.00
312 30 10 50 rep 2.txt	30	10	50	Optimal	0.04	694	694.00	0.00
313 30 10 50 rep 3.txt	30	10	50	Optimal	0.09	762	762.00	0.00
314 30 10 50 rep 4.txt	30	10	50	Optimal	33.03	671	671.00	0.00
315 30 10 50 rep 5.txt	30	10	50	Optimal	0.05	747	747.00	0.00
316 30 10 50 rep 6.txt	30	10	50	Optimal	300.02	664	664.00	0.00
317 30 10 50 rep 7.txt	30	10	50	Optimal	0.53	659	659.00	0.00
318 30 10 50 rep 8.txt	30	10	50	Optimal	0.11	668	668.00	0.00
319 30 10 50 rep 9.txt	30	10	50	Optimal	0.42	658	658.00	0.00
320 30 10 50 rep 10.txt	30	10	50	Optimal	0.03	695	695.00	0.00
321 50 2 10 rep 1.txt	50	2	10	Solution	300.21	1594	1591.00	0.19
322 50 2 10 rep 2.txt	50	2	10	Optimal	300.01	1596	1596.00	0.00
323 50 2 10 rep 3.txt	50	2	10	Solution	300.19	1623	1622.00	0.06
324 50 2 10 rep 4.txt	50	2	10	Optimal	300.03	1819	1819.00	0.00
325 50 2 10 rep 5.txt	50	2	10	Solution	300.16	1671	1654.00	1.02
326 50 2 10 rep 6.txt	50	2	10	Solution	300.12	1289	1278.00	0.85
327 50 2 10 rep 7.txt	50	2	10	Optimal	300.02	1747	1747.00	0.00
328 50 2 10 rep 8.txt	50	2	10	Optimal	300.02	1611	1611.00	0.00
329 50 2 10 rep 9.txt	50	2	10	Solution	300.17	1365	1362.00	0.22
330 50 2 10 rep 10.txt	50	2	10	Solution	300.12	1498	1482.00	1.07
331 50 2 20 rep 1.txt	50	2	20	Solution	300.09	851	836.00	1.76
332 50 2 20 rep 2.txt	50	2	20	Solution	300.10	655	645.00	1.53
333 50 2 20 rep 3.txt	50	2	20	Solution	300.12	770	765.00	0.65
334 50 2 20 rep 4.txt	50	2	20	Solution	300.11	668	663.00	0.75
335 50 2 20 rep 5.txt	50	2	20	Solution	300.15	813	804.00	1.11
336 50 2 20 rep 6.txt	50	2	20	Solution	300.12	810	803.00	0.86
337 50 2 20 rep 7.txt	50	2	20	Solution	300.13	677	661.00	2.36

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
338 50 2 20 rep 8.txt	50	2	20	Solution	300.12	782	766.00	2.05
339 50 2 20 rep 9.txt	50	2	20	Solution	300.13	738	732.00	0.81
340 50 2 20 rep 10.txt	50	2	20	Solution	300.10	684	671.00	1.90
341 50 2 30 rep 1.txt	50	2	30	Solution	300.07	459	450.00	1.96
342 50 2 30 rep 2.txt	50	2	30	Solution	300.07	476	470.00	1.26
343 50 2 30 rep 3.txt	50	2	30	Solution	300.05	466	460.00	1.29
344 50 2 30 rep 4.txt	50	2	30	Solution	300.05	509	502.00	1.38
345 50 2 30 rep 5.txt	50	2	30	Solution	300.09	594	587.00	1.18
346 50 2 30 rep 6.txt	50	2	30	Solution	300.08	544	532.00	2.21
347 50 2 30 rep 7.txt	50	2	30	Solution	300.06	559	544.00	2.68
348 50 2 30 rep 8.txt	50	2	30	Solution	300.09	528	513.00	2.84
349 50 2 30 rep 9.txt	50	2	30	Solution	300.06	479	466.00	2.71
350 50 2 30 rep 10.txt	50	2	30	Solution	300.06	435	423.00	2.76
351 50 2 50 rep 1.txt	50	2	50	Solution	300.04	334	330.00	1.20
352 50 2 50 rep 2.txt	50	2	50	Solution	300.05	286	283.00	1.05
353 50 2 50 rep 3.txt	50	2	50	Solution	300.05	399	393.00	1.50
354 50 2 50 rep 4.txt	50	2	50	Solution	300.05	340	334.00	1.76
355 50 2 50 rep 5.txt	50	2	50	Solution	300.04	297	295.00	0.67
356 50 2 50 rep 6.txt	50	2	50	Solution	300.04	337	331.00	1.78
357 50 2 50 rep 7.txt	50	2	50	Solution	300.06	322	315.00	2.17
358 50 2 50 rep 8.txt	50	2	50	Solution	300.04	325	322.00	0.92
359 50 2 50 rep 9.txt	50	2	50	Solution	300.04	328	323.00	1.52
360 50 2 50 rep 10.txt	50	2	50	Solution	300.03	281	279.00	0.71
361 50 3 10 rep 1.txt	50	3	10	Solution	300.12	1833	1824.00	0.49
362 50 3 10 rep 2.txt	50	3	10	Solution	300.08	1579	1578.00	0.06
363 50 3 10 rep 3.txt	50	3	10	Optimal	300.02	1536	1536.00	0.00
364 50 3 10 rep 4.txt	50	3	10	Solution	300.10	1773	1761.00	0.68
365 50 3 10 rep 5.txt	50	3	10	Solution	300.30	1891	1885.00	0.32
366 50 3 10 rep 6.txt	50	3	10	Solution	300.25	1704	1683.00	1.23
367 50 3 10 rep 7.txt	50	3	10	Solution	300.20	1533	1528.00	0.33
368 50 3 10 rep 8.txt	50	3	10	Solution	300.15	1562	1496.00	4.23
369 50 3 10 rep 9.txt	50	3	10	Solution	300.11	1864	1862.00	0.11
370 50 3 10 rep 10.txt	50	3	10	Solution	300.15	1423	1396.00	1.90
371 50 3 20 rep 1.txt	50	3	20	Solution	300.10	748	725.00	3.07
372 50 3 20 rep 2.txt	50	3	20	Solution	300.07	785	758.00	3.44
373 50 3 20 rep 3.txt	50	3	20	Solution	30.05	791	760.00	3.92
374 50 3 20 rep 4.txt	50	3	20	Solution	30.03	790	775.00	1.90
375 50 3 20 rep 5.txt	50	3	20	Solution	30.04	769	736.00	4.29
376 50 3 20 rep 6.txt	50	3	20	Solution	30.04	872	860.00	1.38
377 50 3 20 rep 7.txt	50	3	20	Solution	30.04	821	800.00	2.56
378 50 3 20 rep 8.txt	50	3	20	Solution	30.04	886	843.00	4.85
379 50 3 20 rep 9.txt	50	3	20	Solution	30.03	963	924.00	4.05
380 50 3 20 rep 10.txt	50	3	20	Solution	30.02	806	755.00	6.33
381 50 3 30 rep 1.txt	50	3	30	Solution	30.02	527	516.00	2.09
382 50 3 30 rep 2.txt	50	3	30	Solution	30.03	507	482.00	4.93

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
383 50 3 30 rep 3.txt	50	3	30	Solution	30.02	516	488.00	5.43
384 50 3 30 rep 4.txt	50	3	30	Solution	30.03	564	536.00	4.96
385 50 3 30 rep 5.txt	50	3	30	Solution	30.01	549	512.00	6.74
386 50 3 30 rep 6.txt	50	3	30	Solution	30.03	589	559.00	5.09
387 50 3 30 rep 7.txt	50	3	30	Solution	30.02	512	491.00	4.10
388 50 3 30 rep 8.txt	50	3	30	Solution	30.03	612	586.00	4.25
389 50 3 30 rep 9.txt	50	3	30	Solution	30.02	523	496.00	5.16
390 50 3 30 rep 10.txt	50	3	30	Solution	30.02	535	514.00	3.93
391 50 3 50 rep 1.txt	50	3	50	Solution	30.03	339	315.00	7.08
392 50 3 50 rep 2.txt	50	3	50	Solution	30.01	301	286.00	4.98
393 50 3 50 rep 3.txt	50	3	50	Solution	30.03	372	362.00	2.69
394 50 3 50 rep 4.txt	50	3	50	Solution	30.02	366	352.00	3.83
395 50 3 50 rep 5.txt	50	3	50	Solution	30.05	385	370.00	3.90
396 50 3 50 rep 6.txt	50	3	50	Solution	30.04	355	348.00	1.97
397 50 3 50 rep 7.txt	50	3	50	Solution	30.01	364	349.00	4.12
398 50 3 50 rep 8.txt	50	3	50	Solution	30.02	362	346.00	4.42
399 50 3 50 rep 9.txt	50	3	50	Solution	30.01	331	322.00	2.72
400 50 3 50 rep 10.txt	50	3	50	Solution	30.03	347	326.00	6.05
401 50 5 10 rep 1.txt	50	5	10	Solution	30.03	1959	1313.00	32.98
402 50 5 10 rep 2.txt	50	5	10	Solution	30.04	1692	1260.00	25.53
403 50 5 10 rep 3.txt	50	5	10	Solution	30.02	1732	1093.00	36.89
404 50 5 10 rep 4.txt	50	5	10	Solution	30.02	2052	1385.00	32.50
405 50 5 10 rep 5.txt	50	5	10	Solution	30.04	1701	1117.00	34.33
406 50 5 10 rep 6.txt	50	5	10	Solution	30.03	1877	1241.00	33.88
407 50 5 10 rep 7.txt	50	5	10	Solution	30.02	1973	1196.00	39.38
408 50 5 10 rep 8.txt	50	5	10	Solution	30.02	1967	1398.00	28.93
409 50 5 10 rep 9.txt	50	5	10	Solution	30.03	1937	1598.00	17.50
410 50 5 10 rep 10.txt	50	5	10	Solution	30.02	1979	1225.00	38.10
411 50 5 20 rep 1.txt	50	5	20	Solution	30.02	900	829.00	7.89
412 50 5 20 rep 2.txt	50	5	20	Solution	30.03	912	871.00	4.50
413 50 5 20 rep 3.txt	50	5	20	Solution	30.05	883	800.00	9.40
414 50 5 20 rep 4.txt	50	5	20	Solution	30.02	936	878.00	6.20
415 50 5 20 rep 5.txt	50	5	20	Solution	30.04	873	806.00	7.67
416 50 5 20 rep 6.txt	50	5	20	Solution	30.06	801	711.00	11.24
417 50 5 20 rep 7.txt	50	5	20	Solution	30.03	951	850.00	10.62
418 50 5 20 rep 8.txt	50	5	20	Solution	30.02	852	806.00	5.40
419 50 5 20 rep 9.txt	50	5	20	Solution	30.02	932	826.00	11.37
420 50 5 20 rep 10.txt	50	5	20	Solution	30.04	902	789.00	12.53
421 50 5 30 rep 1.txt	50	5	30	Solution	30.03	680	614.00	9.71
422 50 5 30 rep 2.txt	50	5	30	Solution	30.02	700	657.00	6.14
423 50 5 30 rep 3.txt	50	5	30	Solution	30.03	697	660.00	5.31
424 50 5 30 rep 4.txt	50	5	30	Solution	30.02	618	576.00	6.80
425 50 5 30 rep 5.txt	50	5	30	Solution	30.02	656	628.00	4.27
426 50 5 30 rep 6.txt	50	5	30	Solution	30.03	677	628.00	7.24
427 50 5 30 rep 7.txt	50	5	30	Solution	30.02	673	623.00	7.43

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
428 50 5 30 rep 8.txt	50	5	30	Solution	30.03	732	700.00	4.37
429 50 5 30 rep 9.txt	50	5	30	Solution	30.02	658	619.00	5.93
430 50 5 30 rep 10.txt	50	5	30	Solution	30.04	680	660.00	2.94
431 50 5 50 rep 1.txt	50	5	50	Solution	30.02	453	415.00	8.39
432 50 5 50 rep 2.txt	50	5	50	Solution	30.07	475	448.00	5.68
433 50 5 50 rep 3.txt	50	5	50	Solution	30.02	458	444.00	3.06
434 50 5 50 rep 4.txt	50	5	50	Solution	30.01	492	449.00	8.74
435 50 5 50 rep 5.txt	50	5	50	Solution	30.04	471	439.00	6.79
436 50 5 50 rep 6.txt	50	5	50	Solution	30.02	486	461.00	5.14
437 50 5 50 rep 7.txt	50	5	50	Solution	30.03	460	427.00	7.17
438 50 5 50 rep 8.txt	50	5	50	Solution	30.02	479	443.00	7.52
439 50 5 50 rep 9.txt	50	5	50	Solution	30.04	510	471.00	7.65
440 50 5 50 rep 10.txt	50	5	50	Solution	30.03	477	432.00	9.43
441 50 10 10 rep 1.txt	50	10	10	Solution	30.03	2524	1217.00	51.78
442 50 10 10 rep 2.txt	50	10	10	Solution	30.04	2401	1288.00	46.36
443 50 10 10 rep 3.txt	50	10	10	Solution	30.04	2580	1380.00	46.51
444 50 10 10 rep 4.txt	50	10	10	Solution	30.04	2693	1392.00	48.31
445 50 10 10 rep 5.txt	50	10	10	Solution	30.04	2315	1292.00	44.19
446 50 10 10 rep 6.txt	50	10	10	Solution	30.05	2545	1311.00	48.49
447 50 10 10 rep 7.txt	50	10	10	Solution	30.04	2465	1470.00	40.37
448 50 10 10 rep 8.txt	50	10	10	Solution	30.04	2447	1350.00	44.83
449 50 10 10 rep 9.txt	50	10	10	Solution	30.03	2461	1309.00	46.81
450 50 10 10 rep 10.txt	50	10	10	Solution	30.05	2469	1316.00	46.70
451 50 10 20 rep 1.txt	50	10	20	Solution	30.08	1342	1088.00	18.93
452 50 10 20 rep 2.txt	50	10	20	Solution	30.06	1340	1019.00	23.96
453 50 10 20 rep 3.txt	50	10	20	Solution	30.04	1316	931.00	29.26
454 50 10 20 rep 4.txt	50	10	20	Solution	30.34	1376	1060.00	22.97
455 50 10 20 rep 5.txt	50	10	20	Solution	30.03	1339	1003.00	25.09
456 50 10 20 rep 6.txt	50	10	20	Solution	30.35	1295	1048.00	19.07
457 50 10 20 rep 7.txt	50	10	20	Solution	30.06	1373	1133.00	17.48
458 50 10 20 rep 8.txt	50	10	20	Solution	30.05	1303	1097.00	15.81
459 50 10 20 rep 9.txt	50	10	20	Solution	30.03	1288	1116.00	13.35
460 50 10 20 rep 10.txt	50	10	20	Solution	30.04	1375	1161.00	15.56
461 50 10 30 rep 1.txt	50	10	30	Solution	30.04	1016	864.00	14.96
462 50 10 30 rep 2.txt	50	10	30	Solution	30.03	936	883.00	5.66
463 50 10 30 rep 3.txt	50	10	30	Solution	30.05	953	784.00	17.73
464 50 10 30 rep 4.txt	50	10	30	Solution	30.04	946	791.00	16.38
465 50 10 30 rep 5.txt	50	10	30	Solution	30.26	981	889.00	9.38
466 50 10 30 rep 6.txt	50	10	30	Solution	30.03	976	873.00	10.55
467 50 10 30 rep 7.txt	50	10	30	Solution	30.03	1014	886.00	12.62
468 50 10 30 rep 8.txt	50	10	30	Solution	30.06	944	829.00	12.18
469 50 10 30 rep 9.txt	50	10	30	Solution	30.03	1021	856.00	16.16
470 50 10 30 rep 10.txt	50	10	30	Solution	30.04	1038	910.00	12.33
471 50 10 50 rep 1.txt	50	10	50	Optimal	30.01	727	727.00	0.00
472 50 10 50 rep 2.txt	50	10	50	Solution	30.03	702	695.00	1.00

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
473 50 10 50 rep 3.txt	50	10	50	Solution	30.03	693	651.00	6.06
474 50 10 50 rep 4.txt	50	10	50	Solution	30.03	759	720.00	5.14
475 50 10 50 rep 5.txt	50	10	50	Solution	30.03	715	696.00	2.66
476 50 10 50 rep 6.txt	50	10	50	Solution	30.03	770	751.00	2.47
477 50 10 50 rep 7.txt	50	10	50	Solution	30.03	758	706.00	6.86
478 50 10 50 rep 8.txt	50	10	50	Solution	30.03	726	714.00	1.65
479 50 10 50 rep 9.txt	50	10	50	Optimal	30.01	767	767.00	0.00
480 50 10 50 rep 10.txt	50	10	50	Solution	30.03	700	661.00	5.57
481 100 2 10 rep 1.txt	100	2	10	Solution	30.05	2625	1835.00	30.10
482 100 2 10 rep 2.txt	100	2	10	Solution	30.07	3104	1721.00	44.56
483 100 2 10 rep 3.txt	100	2	10	Solution	30.03	2809	1644.00	41.47
484 100 2 10 rep 4.txt	100	2	10	Solution	30.04	2871	2062.00	28.18
485 100 2 10 rep 5.txt	100	2	10	Solution	30.09	3033	1724.00	43.16
486 100 2 10 rep 6.txt	100	2	10	Solution	30.03	3324	1844.00	44.52
487 100 2 10 rep 7.txt	100	2	10	Solution	30.02	2971	1571.00	47.12
488 100 2 10 rep 8.txt	100	2	10	Solution	30.03	3141	1632.00	48.04
489 100 2 10 rep 9.txt	100	2	10	Solution	30.03	2997	1708.00	43.01
490 100 2 10 rep 10.txt	100	2	10	Solution	30.03	2998	1715.00	42.80
491 100 2 20 rep 1.txt	100	2	20	Solution	30.02	1458	943.00	35.32
492 100 2 20 rep 2.txt	100	2	20	Solution	30.03	1522	1361.00	10.58
493 100 2 20 rep 3.txt	100	2	20	Solution	30.03	1337	958.00	28.35
494 100 2 20 rep 4.txt	100	2	20	Solution	30.25	1349	1263.00	6.38
495 100 2 20 rep 5.txt	100	2	20	Solution	30.07	1418	1378.00	2.82
496 100 2 20 rep 6.txt	100	2	20	Solution	30.03	1485	928.00	37.51
497 100 2 20 rep 7.txt	100	2	20	Solution	30.02	1467	1003.00	31.63
498 100 2 20 rep 8.txt	100	2	20	Solution	30.04	1536	1047.00	31.84
499 100 2 20 rep 9.txt	100	2	20	Solution	30.03	1363	1232.00	9.61
500 100 2 20 rep 10.txt	100	2	20	Solution	30.04	1495	1010.00	32.44
501 100 2 30 rep 1.txt	100	2	30	Solution	30.03	951	917.00	3.58
502 100 2 30 rep 2.txt	100	2	30	Solution	30.03	882	799.00	9.41
503 100 2 30 rep 3.txt	100	2	30	Solution	30.03	1042	866.00	16.89
504 100 2 30 rep 4.txt	100	2	30	Solution	30.02	1008	832.00	17.46
505 100 2 30 rep 5.txt	100	2	30	Solution	30.03	966	692.00	28.36
506 100 2 30 rep 6.txt	100	2	30	Solution	30.02	941	910.00	3.29
507 100 2 30 rep 7.txt	100	2	30	Solution	30.03	992	781.00	21.27
508 100 2 30 rep 8.txt	100	2	30	Solution	30.04	922	882.00	4.34
509 100 2 30 rep 9.txt	100	2	30	Solution	30.02	921	892.00	3.15
510 100 2 30 rep 10.txt	100	2	30	Solution	30.04	1014	770.00	24.06
511 100 2 50 rep 1.txt	100	2	50	Solution	30.03	667	646.00	3.15
512 100 2 50 rep 2.txt	100	2	50	Solution	30.03	584	560.00	4.11
513 100 2 50 rep 3.txt	100	2	50	Solution	30.02	581	565.00	2.75
514 100 2 50 rep 4.txt	100	2	50	Solution	30.03	621	609.00	1.93
515 100 2 50 rep 5.txt	100	2	50	Solution	30.02	581	567.00	2.41
516 100 2 50 rep 6.txt	100	2	50	Solution	30.03	697	685.00	1.72
517 100 2 50 rep 7.txt	100	2	50	Solution	30.04	598	580.00	3.01

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
518 100 2 50 rep 8.txt	100	2	50	Solution	30.03	626	616.00	1.60
519 100 2 50 rep 9.txt	100	2	50	Solution	30.04	595	582.00	2.18
520 100 2 50 rep 10.txt	100	2	50	Solution	30.02	549	540.00	1.64
521 100 3 10 rep 1.txt	100	3	10	Solution	30.05	3005	1611.00	46.39
522 100 3 10 rep 2.txt	100	3	10	Solution	30.03	3158	1704.00	46.04
523 100 3 10 rep 3.txt	100	3	10	Solution	30.02	3671	1889.00	48.54
524 100 3 10 rep 4.txt	100	3	10	Solution	30.04	3113	1572.00	49.50
525 100 3 10 rep 5.txt	100	3	10	Solution	30.02	3150	1666.00	47.11
526 100 3 10 rep 6.txt	100	3	10	Solution	30.03	3447	1832.00	46.85
527 100 3 10 rep 7.txt	100	3	10	Solution	30.02	3309	1619.00	51.07
528 100 3 10 rep 8.txt	100	3	10	Solution	30.06	3341	1648.00	50.67
529 100 3 10 rep 9.txt	100	3	10	Solution	30.05	3375	1646.00	51.23
530 100 3 10 rep 10.txt	100	3	10	Solution	30.03	3274	1704.00	47.95
531 100 3 20 rep 1.txt	100	3	20	Solution	30.04	1608	998.00	37.94
532 100 3 20 rep 2.txt	100	3	20	Solution	30.03	1621	1000.00	38.31
533 100 3 20 rep 3.txt	100	3	20	Solution	30.03	1499	1131.00	24.55
534 100 3 20 rep 4.txt	100	3	20	Solution	30.04	1595	930.00	41.69
535 100 3 20 rep 5.txt	100	3	20	Solution	30.04	1560	1025.00	34.29
536 100 3 20 rep 6.txt	100	3	20	Solution	30.03	1583	1108.00	30.01
537 100 3 20 rep 7.txt	100	3	20	Solution	30.04	1599	944.00	40.96
538 100 3 20 rep 8.txt	100	3	20	Solution	30.02	1648	1055.00	35.98
539 100 3 20 rep 9.txt	100	3	20	Solution	30.05	1473	1148.00	22.06
540 100 3 20 rep 10.txt	100	3	20	Solution	30.04	1789	1018.00	43.10
541 100 3 30 rep 1.txt	100	3	30	Solution	30.03	1038	772.00	25.63
542 100 3 30 rep 2.txt	100	3	30	Solution	30.04	1091	749.00	31.35
543 100 3 30 rep 3.txt	100	3	30	Solution	30.02	1087	783.00	27.97
544 100 3 30 rep 4.txt	100	3	30	Solution	30.03	1080	775.00	28.24
545 100 3 30 rep 5.txt	100	3	30	Solution	30.03	1180	786.00	33.39
546 100 3 30 rep 6.txt	100	3	30	Solution	30.04	1014	785.00	22.58
547 100 3 30 rep 7.txt	100	3	30	Solution	30.04	1088	819.00	24.72
548 100 3 30 rep 8.txt	100	3	30	Solution	30.07	1059	876.00	17.28
549 100 3 30 rep 9.txt	100	3	30	Solution	30.04	950	749.00	21.16
550 100 3 30 rep 10.txt	100	3	30	Solution	30.02	1159	854.00	26.32
551 100 3 50 rep 1.txt	100	3	50	Solution	30.04	618	576.00	6.80
552 100 3 50 rep 2.txt	100	3	50	Solution	30.03	700	668.00	4.57
553 100 3 50 rep 3.txt	100	3	50	Solution	30.03	661	625.00	5.45
554 100 3 50 rep 4.txt	100	3	50	Solution	30.04	655	625.00	4.58
555 100 3 50 rep 5.txt	100	3	50	Solution	30.03	652	600.00	7.98
556 100 3 50 rep 6.txt	100	3	50	Solution	30.04	652	614.00	5.83
557 100 3 50 rep 7.txt	100	3	50	Solution	30.03	733	693.00	5.46
558 100 3 50 rep 8.txt	100	3	50	Solution	30.02	653	617.00	5.51
559 100 3 50 rep 9.txt	100	3	50	Solution	30.04	604	586.00	2.98
560 100 3 50 rep 10.txt	100	3	50	Solution	30.03	651	614.00	5.68
561 100 5 10 rep 1.txt	100	5	10	Solution	30.03	3448	1601.00	53.57
562 100 5 10 rep 2.txt	100	5	10	Solution	30.12	3591	1611.00	55.14

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
563 100 5 10 rep 3.txt	100	5	10	Solution	30.04	3870	1734.00	55.19
564 100 5 10 rep 4.txt	100	5	10	Solution	30.22	4106	1806.00	56.02
565 100 5 10 rep 5.txt	100	5	10	Solution	30.04	3598	1655.00	54.00
566 100 5 10 rep 6.txt	100	5	10	Solution	30.03	3606	1760.00	51.19
567 100 5 10 rep 7.txt	100	5	10	Solution	30.04	3990	1807.00	54.71
568 100 5 10 rep 8.txt	100	5	10	Solution	30.03	3992	1841.00	53.88
569 100 5 10 rep 9.txt	100	5	10	Solution	30.03	3671	1697.00	53.77
570 100 5 10 rep 10.txt	100	5	10	Solution	30.03	3667	1651.00	54.98
571 100 5 20 rep 1.txt	100	5	20	Solution	30.21	1745	1034.00	40.74
572 100 5 20 rep 2.txt	100	5	20	Solution	30.04	1908	1021.00	46.49
573 100 5 20 rep 3.txt	100	5	20	Solution	30.05	1914	988.00	48.38
574 100 5 20 rep 4.txt	100	5	20	Solution	30.03	1757	1000.00	43.08
575 100 5 20 rep 5.txt	100	5	20	Solution	30.04	1830	1159.00	36.67
576 100 5 20 rep 6.txt	100	5	20	Solution	30.04	1822	1114.00	38.86
577 100 5 20 rep 7.txt	100	5	20	Solution	30.03	1786	1130.00	36.73
578 100 5 20 rep 8.txt	100	5	20	Solution	30.04	1809	996.00	44.94
579 100 5 20 rep 9.txt	100	5	20	Solution	30.04	1673	1071.00	35.98
580 100 5 20 rep 10.txt	100	5	20	Solution	30.04	1693	1107.00	34.61
581 100 5 30 rep 1.txt	100	5	30	Solution	30.18	1152	847.00	26.48
582 100 5 30 rep 2.txt	100	5	30	Solution	30.04	1217	777.00	36.15
583 100 5 30 rep 3.txt	100	5	30	Solution	30.31	1151	783.00	31.97
584 100 5 30 rep 4.txt	100	5	30	Solution	30.04	1185	871.00	26.50
585 100 5 30 rep 5.txt	100	5	30	Solution	30.05	1213	737.00	39.24
586 100 5 30 rep 6.txt	100	5	30	Solution	30.06	1224	771.00	37.01
587 100 5 30 rep 7.txt	100	5	30	Solution	30.04	1265	794.00	37.23
588 100 5 30 rep 8.txt	100	5	30	Solution	30.03	1211	798.00	34.10
589 100 5 30 rep 9.txt	100	5	30	Solution	30.14	1229	794.00	35.39
590 100 5 30 rep 10.txt	100	5	30	Solution	30.03	1362	1027.00	24.60
591 100 5 50 rep 1.txt	100	5	50	Solution	30.03	802	720.00	10.22
592 100 5 50 rep 2.txt	100	5	50	Solution	30.04	744	666.00	10.48
593 100 5 50 rep 3.txt	100	5	50	Solution	30.04	813	737.00	9.35
594 100 5 50 rep 4.txt	100	5	50	Solution	30.05	836	690.00	17.46
595 100 5 50 rep 5.txt	100	5	50	Solution	30.04	812	699.00	13.92
596 100 5 50 rep 6.txt	100	5	50	Solution	30.04	822	765.00	6.93
597 100 5 50 rep 7.txt	100	5	50	Solution	30.03	765	706.00	7.71
598 100 5 50 rep 8.txt	100	5	50	Solution	30.04	808	727.00	10.02
599 100 5 50 rep 9.txt	100	5	50	Solution	30.08	810	715.00	11.73
600 100 5 50 rep 10.txt	100	5	50	Solution	30.04	739	690.00	6.63
601 100 10 10 rep 1.txt	100	10	10	Solution	30.06	4495	2041.00	54.59
602 100 10 10 rep 2.txt	100	10	10	Solution	30.04	4676	1985.00	57.55
603 100 10 10 rep 3.txt	100	10	10	Solution	30.03	4328	1871.00	56.77
604 100 10 10 rep 4.txt	100	10	10	Solution	30.04	4702	2020.00	57.04
605 100 10 10 rep 5.txt	100	10	10	Solution	30.04	4739	2129.00	55.07
606 100 10 10 rep 6.txt	100	10	10	Solution	30.04	4743	3490.00	26.42
607 100 10 10 rep 7.txt	100	10	10	Solution	30.04	4609	2001.00	56.58

Table 2: Results for CFS (CPSat) (640 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
608 100 10 10 rep 8.txt	100	10	10	Solution	30.04	4333	1856.00	57.17
609 100 10 10 rep 9.txt	100	10	10	Solution	30.04	4627	1892.00	59.11
610 100 10 10 rep 10.txt	100	10	10	Solution	30.05	4710	1918.00	59.28
611 100 10 20 rep 1.txt	100	10	20	Solution	30.04	2245	1183.00	47.31
612 100 10 20 rep 2.txt	100	10	20	Solution	30.04	2329	1138.00	51.14
613 100 10 20 rep 3.txt	100	10	20	Solution	30.51	2388	1151.00	51.80
614 100 10 20 rep 4.txt	100	10	20	Solution	30.04	2318	1210.00	47.80
615 100 10 20 rep 5.txt	100	10	20	Solution	30.07	2312	1186.00	48.70
616 100 10 20 rep 6.txt	100	10	20	Solution	30.04	2430	1168.00	51.93
617 100 10 20 rep 7.txt	100	10	20	Solution	30.03	2255	1116.00	50.51
618 100 10 20 rep 8.txt	100	10	20	Solution	30.05	2220	1153.00	48.06
619 100 10 20 rep 9.txt	100	10	20	Solution	30.03	2207	1118.00	49.34
620 100 10 20 rep 10.txt	100	10	20	Solution	30.04	2186	1228.00	43.82
621 100 10 30 rep 1.txt	100	10	30	Solution	30.06	1652	913.00	44.73
622 100 10 30 rep 2.txt	100	10	30	Solution	30.04	1619	943.00	41.75
623 100 10 30 rep 3.txt	100	10	30	Solution	30.06	1490	976.00	34.50
624 100 10 30 rep 4.txt	100	10	30	Solution	30.05	1616	931.00	42.39
625 100 10 30 rep 5.txt	100	10	30	Solution	30.03	1614	941.00	41.70
626 100 10 30 rep 6.txt	100	10	30	Solution	30.33	1654	966.00	41.60
627 100 10 30 rep 7.txt	100	10	30	Solution	30.05	1568	962.00	38.65
628 100 10 30 rep 8.txt	100	10	30	Solution	30.04	1690	941.00	44.32
629 100 10 30 rep 9.txt	100	10	30	Solution	30.03	1670	962.00	42.40
630 100 10 30 rep 10.txt	100	10	30	Solution	30.06	1722	954.00	44.60
631 100 10 50 rep 1.txt	100	10	50	Solution	30.04	1188	996.00	16.16
632 100 10 50 rep 2.txt	100	10	50	Solution	30.05	1143	870.00	23.88
633 100 10 50 rep 3.txt	100	10	50	Solution	30.08	1108	901.00	18.68
634 100 10 50 rep 4.txt	100	10	50	Solution	30.05	1036	845.00	18.44
635 100 10 50 rep 5.txt	100	10	50	Solution	30.09	1067	878.00	17.71
636 100 10 50 rep 6.txt	100	10	50	Solution	30.04	1128	940.00	16.67
637 100 10 50 rep 7.txt	100	10	50	Solution	30.05	1125	868.00	22.84
638 100 10 50 rep 8.txt	100	10	50	Solution	30.08	1097	916.00	16.50
639 100 10 50 rep 9.txt	100	10	50	Solution	30.05	1104	851.00	22.92
640 100 10 50 rep 10.txt	100	10	50	Solution	30.06	1170	896.00	23.42

3 Comparison

Table 3: Comparison of CPO and CPSat for Results of CFS prec

Group	Nr	All Instances				Optimal Only		Non Optimal Only			
		Optimal (% of All Instances)				Time (% of VB)		Cost (% of VB)		Bound (% of VB)	
		Both	CPO	CPSat	None	CPO	CPSat	CPO	CPSat	CPO	CPSat
20/0/2/10	10	10.00	0.00	40.00	50.00	100.00	289.44	100.00	100.00	91.34	100.00
20/0/2/20	10	80.00	20.00	0.00	0.00	100.00	5043.68	100.00	100.00	100.00	98.76
20/0/2/30	10	100.00	0.00	0.00	0.00	100.14	4195.18	n/a	n/a	n/a	n/a
20/0/2/50	10	100.00	0.00	0.00	0.00	100.00	38905.21	n/a	n/a	n/a	n/a
20/0/3/10	10	0.00	0.00	70.00	30.00	n/a	n/a	100.18	100.00	89.65	100.00
20/0/3/20	10	50.00	0.00	20.00	30.00	111.94	1738.13	100.51	100.00	98.46	99.74
20/0/3/30	10	100.00	0.00	0.00	0.00	100.52	8057.85	n/a	n/a	n/a	n/a
20/0/3/50	10	100.00	0.00	0.00	0.00	100.00	36779.23	n/a	n/a	n/a	n/a
20/0/5/10	10	0.00	0.00	30.00	70.00	n/a	n/a	100.68	100.00	91.67	100.00
20/0/5/20	10	10.00	0.00	20.00	70.00	100.00	5900.24	101.00	100.04	98.07	99.52
20/0/5/30	10	90.00	0.00	0.00	10.00	100.00	9933.68	100.78	100.00	98.15	100.00
20/0/5/50	10	100.00	0.00	0.00	0.00	100.00	212.63	n/a	n/a	n/a	n/a
20/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	103.11	100.00	91.70	100.00
20/0/10/20	10	0.00	0.00	10.00	90.00	n/a	n/a	101.47	100.00	97.58	99.91
20/0/10/30	10	90.00	0.00	10.00	0.00	100.01	1523.08	101.09	100.00	99.69	100.00
20/0/10/50	10	100.00	0.00	0.00	0.00	101.13	136.95	n/a	n/a	n/a	n/a
30/0/2/10	10	0.00	0.00	30.00	70.00	n/a	n/a	100.18	100.00	94.02	100.00
30/0/2/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.51	100.16	99.75	100.00
30/0/2/30	10	10.00	0.00	0.00	90.00	100.00	3131.70	100.26	100.03	99.97	100.00
30/0/2/50	10	60.00	0.00	10.00	30.00	100.00	6656.70	100.46	100.00	99.65	99.77
30/0/3/10	10	0.00	0.00	40.00	60.00	n/a	n/a	100.31	100.04	92.94	100.00
30/0/3/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.83	100.13	98.93	100.00
30/0/3/30	10	0.00	0.00	10.00	90.00	n/a	n/a	100.39	100.14	99.72	100.00
30/0/3/50	10	90.00	0.00	0.00	10.00	101.51	4072.70	100.39	100.00	100.00	100.00
30/0/5/10	10	0.00	0.00	10.00	90.00	n/a	n/a	100.60	100.42	94.57	99.99
30/0/5/20	10	0.00	0.00	0.00	100.00	n/a	n/a	101.43	100.39	98.60	100.00
30/0/5/30	10	0.00	0.00	10.00	90.00	n/a	n/a	100.90	100.04	99.42	99.96
30/0/5/50	10	80.00	0.00	10.00	10.00	100.00	115540.40	100.63	100.00	99.50	97.86
30/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	101.21	100.63	91.22	100.00
30/0/10/20	10	0.00	0.00	0.00	100.00	n/a	n/a	101.32	100.40	98.79	99.96
30/0/10/30	10	10.00	0.00	0.00	90.00	100.00	48388.27	101.24	100.11	99.49	99.97
30/0/10/50	10	90.00	0.00	10.00	0.00	121.58	3876.48	100.45	100.00	98.34	100.00
50/0/2/10	10	0.00	0.00	40.00	60.00	n/a	n/a	100.09	100.01	94.29	100.00
50/0/2/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.34	100.12	99.96	100.00
50/0/2/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.32	100.34	99.78	99.98
50/0/2/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.15	100.25	99.94	100.00
50/0/3/10	10	0.00	0.00	10.00	90.00	n/a	n/a	100.06	100.31	93.84	100.00
50/0/3/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.48	100.80	99.89	99.99
50/0/3/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	101.36	99.77	99.92
50/0/3/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.06	101.15	99.59	99.97
50/0/5/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.06	101.56	100.00	75.84
50/0/5/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.02	102.36	99.52	99.98
50/0/5/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.09	101.93	99.50	99.95
50/0/5/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.13	101.62	100.00	99.80
50/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	106.21	100.00	69.80
50/0/10/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	106.01	99.80	96.35
50/0/10/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	103.05	99.59	99.95
50/0/10/50	10	20.00	10.00	0.00	70.00	100.00	1869.85	100.00	101.32	100.00	99.95
100/0/2/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	101.82	100.00	61.59
100/0/2/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	101.80	100.00	79.95
100/0/2/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	101.78	99.94	89.39
100/0/2/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	101.09	99.68	100.00
100/0/3/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	103.74	100.00	56.81
100/0/3/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.01	102.50	100.00	68.63
100/0/3/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	103.78	100.00	79.19
100/0/3/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	103.05	99.58	99.98
100/0/5/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	106.29	100.00	54.55
100/0/5/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	105.96	100.00	66.62
100/0/5/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	106.66	100.00	77.20
100/0/5/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	105.97	99.33	99.45
100/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	109.82	99.98	62.44
100/0/10/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	108.04	100.00	65.43
100/0/10/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	109.22	100.00	73.38
100/0/10/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.00	108.94	99.69	98.15